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WAFER MOUNTING APPARATUS, SEMICONDUCTOR MANUFACTURING APPARATUS, METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE, AND SEMICONDUCTOR DEVICE

Abstract

A ring is mounted on a satellite disk in such a manner that a terrace portion is supported by a peripheral portion front surface of a peripheral portion of the satellite disk. A SiC wafer is mounted on three lift-up pins mounted on a front surface of the terrace portion. A disk upper space is formed between a front surface of an upper layer portion of the satellite disk and a back surface of the SiC wafer. A wall portion of the ring is disposed to face an orientation flat portion via a gap space. On the back surface side of the SiC wafer, an annular groove is formed along an outer periphery of an opening in plan view, and one side surface of the groove is a side surface of the upper layer portion.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present disclosure relates to a wafer mounting apparatus for mounting a semiconductor wafer, typified by a susceptor used for epitaxial growth of a SiC film on a SiC single crystal wafer.

Description of the Background Art

[0002] A susceptor used for epitaxial growth of a SiC film on a SiC single crystal wafer is generally used as a conventional wafer mounting apparatus. A conventional susceptor is disclosed in, for example, Japanese Patent Application Laid-Open No. 2016-119472.

[0003] Japanese Patent Application Laid-Open No. 2016-119472 is characterized by a technique in which a step between an upper surface of a susceptor and an upper surface of a silicon wafer is set to 1 mm or more, and by using a susceptor having this characteristic, an epi-crown phenomenon described later is suppressed. Further, Japanese Patent Application Laid-Open No. 2014-27006 or Japanese Patent Application Laid-Open No. 7-226349 (1995) also discloses a conventional technique for suppressing the epi-crown phenomenon.

[0004] As a technique developed from a wafer mounting apparatus typified by a susceptor, a SiC epitaxial growth apparatus, which is a semiconductor manufacturing apparatus including the wafer mounting apparatus, can be considered. Furthermore, a method of manufacturing a semiconductor device using a SiC epitaxial growth apparatus and a semiconductor device manufactured by the method of manufacturing a semiconductor device are conceivable.

[0005] When an epitaxial layer is formed on a front surface of a semiconductor wafer such as a silicon wafer, there is a wafer susceptor top distance indicating a length in a height direction between the upper surface of the susceptor and the front surface of the semiconductor wafer as an index related to the film thickness of the epitaxial layer. When the wafer susceptor top distance is set to be relatively large, the epi thickness, which is the film thickness of the epitaxial layer, is affected in a peripheral region to be an outermost peripheral portion of the semiconductor wafer, and in-plane uniformity of the epi thickness is deteriorated.

SUMMARY

[0006] The present disclosure has been made to solve the above-described problems, and an object of the present disclosure is to obtain a wafer mounting apparatus that improves in-plane uniformity in an epitaxial layer formed on a front surface of a semiconductor wafer.

[0007] A wafer mounting apparatus according to the present disclosure is a wafer mounting apparatus that mounts a semiconductor wafer having a cutout portion, and includes a disk, a ring, and a plurality of lift-up members.

[0008] The disk has a convex upper layer portion.

[0009] The ring is disposed on the disk and has an opening in the center.

[0010] The plurality of lift-up members is arranged on a lift-up region provided on the disk or the ring, and support the semiconductor wafer from a back surface.

[0011] The ring has a cutout-side protruding wall portion protruding toward the opening.

[0012] A semiconductor wafer mounting state is a state in which the semiconductor wafer is mounted on the plurality of lift-up members so that the semiconductor wafer fits in the opening in plan view without being in contact with the ring.

[0013] In the wafer mounting state, the cutout-side protruding wall portion faces the cutout portion of the semiconductor wafer, and a disk upper space is formed between a front surface of the upper layer portion of the disk and a back surface of the semiconductor wafer except for a formation region of the plurality of lift-up members.

[0014] In the wafer mounting state, an annular groove having an annular shape is formed along an outer periphery of the opening in plan view between the upper layer portion of the disk and the ring on a side of the back surface of the semiconductor wafer.

[0015] In the wafer mounting state, a side surface of the upper layer portion of the disk is a side surface of the annular groove.

[0016] The wafer mounting apparatus of the present disclosure can provide a disk upper space between the back surface of the semiconductor wafer and the front surface of the disk during the semiconductor wafer mounting state.

[0017] Furthermore, in the wafer mounting apparatus of the present disclosure, during the semiconductor wafer mounting state, an annular groove having an annular shape is formed between the upper layer portion of the disk and the ring on the back surface side of the semiconductor wafer, and a side surface of the upper layer portion of the disk is a side surface of the annular groove.

[0018] Therefore, when the source gas is supplied above the front surface of the semiconductor wafer in the semiconductor wafer mounting state, a part of the source gas can be intentionally released to the disk upper space on the back surface side of the semiconductor wafer via the annular groove.

[0019] Therefore, when an epitaxial layer is formed on the front surface of the semiconductor wafer in the semiconductor wafer mounting state by an epitaxial growth process, it is possible to suppress the occurrence of the epi-crown phenomenon in which the epi thickness, which is the film thickness of the epitaxial growth layer, locally increases.

[0020] As a result, the wafer mounting apparatus of the present disclosure has an effect of being able to form an epitaxial layer having an epi thickness with excellent in-plane uniformity on the front surface of the semiconductor wafer in the semiconductor wafer mounting state.

[0021] These and other objects, features, aspects and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is an explanatory view schematically illustrating a planar configuration of a susceptor which is a basic configuration of a first preferred embodiment;

[0023] FIG. 2 is an explanatory view schematically illustrating a configuration of an A1-A1 cross section of FIG. 1;

[0024] FIG. 3 is an explanatory view schematically illustrating a configuration of a B1-B1 cross section of FIG. 1;

[0025] FIG. 4 is an explanatory view schematically illustrating a configuration of a C1-C1 cross section in FIG. 1;

[0026] FIG. 5 is an explanatory view illustrating an enlarged cross-sectional configuration of FIG. 2;

[0027] FIG. 6 is an explanatory view schematically illustrating a planar configuration of a susceptor which is a first modification of the first preferred embodiment;

[0028] FIG. 7 is an explanatory view schematically illustrating a configuration of a cross section taken along line A11-A11 in FIG. 6;

[0029] FIG. 8 is an explanatory view schematically illustrating a configuration of a cross section

taken along line B11-B11 in FIG. 6;

[0030] FIG. 9 is an explanatory view schematically illustrating a configuration of a cross section taken along line C11 to C11 in FIG. 6;

[0031] FIG. 10 is an explanatory view schematically illustrating a planar configuration of a susceptor which is a second modification of the first preferred embodiment;

[0032] FIG. 11 is an explanatory view schematically illustrating a configuration of a cross section taken along line A12-A12 in FIG. 10;

[0033] FIG. 12 is an explanatory view schematically illustrating a configuration of a cross section taken along line B12-B12 in FIG. 10;

[0034] FIG. 13 is an explanatory view schematically illustrating a configuration of a cross section taken along line C12-C12 in FIG. 10;

[0035] FIG. 14 is an explanatory view schematically illustrating a planar configuration of a susceptor which is a basic configuration of a second preferred embodiment;

[0036] FIG. 15 is an explanatory view schematically illustrating a configuration of an A2-A2 cross section in FIG. 14;

[0037] FIG. 16 is an explanatory view schematically illustrating a configuration of a B2-B2 cross section of FIG. 14;

[0038] FIG. 17 is an enlarged explanatory view illustrating a focused region R2 in FIG. 14;

[0039] FIG. 18 is an explanatory view schematically illustrating a planar configuration of a susceptor as a basic configuration of a third preferred embodiment;

[0040] FIG. 19 is an explanatory view schematically illustrating a configuration of an A3-A3 cross section of FIG. 18;

[0041] FIG. 20 is an explanatory view schematically illustrating a configuration of a B4-B4 cross section in FIG. 18;

[0042] FIG. 21 is an explanatory view schematically illustrating a configuration of a C4-C4 cross section in FIG. 18;

[0043] FIG. 22 is an explanatory view schematically illustrating a planar configuration of a susceptor which is a basic configuration of a fourth preferred embodiment;

[0044] FIG. 23 is an explanatory view schematically illustrating a configuration of an A4-A4 cross section in FIG. 22;

[0045] FIG. 24 is an explanatory view schematically illustrating a configuration of a B4-B4 cross section in FIG. 22;

[0046] FIG. 25 is an explanatory view schematically illustrating a cross-sectional structure of a semiconductor manufacturing apparatus including a susceptor used in a preferred embodiment of the present disclosure;

[0047] FIG. 26 is an explanatory view schematically illustrating a planar configuration of the semiconductor manufacturing apparatus;

[0048] FIG. 27 is a flowchart illustrating a processing procedure of a method of manufacturing a semiconductor device using a semiconductor manufacturing apparatus;

[0049] FIG. 28 is a cross-sectional view illustrating a cross-sectional structure of a semiconductor device manufactured by the method of manufacturing a semiconductor device;

[0050] FIG. 29 is an explanatory view schematically illustrating an epi-crown phenomenon;

[0051] FIGS. 30A and 30B are explanatory views schematically illustrating a conventional susceptor;

[0052] FIG. 31 is a graph illustrating an epi-crown height when an epitaxial layer is grown using the susceptor illustrated in FIGS. 30A and 30B;

[0053] FIG. 32 is a diagram schematically illustrating a relationship between a measurement point and a reference point on a front surface of a SiC wafer;

[0054] FIG. 33 is a graph illustrating the correlation of a height of the epi-crown at an OF center portion when a wafer susceptor top distance is changed;

[0055] FIG. **34** is a graph illustrating a correlation between an average epi thickness of a wafer outer peripheral portion and the wafer susceptor top distance;

[0056] FIG. **35** is a graph illustrating an epi thickness when a wafer susceptor top distance is 0.7 mm;

[0057] FIG. **36** is a graph illustrating an epi thickness when a wafer susceptor top distance is 2.5 mm;

[0058] FIG. **37** is an explanatory view schematically illustrating a planar configuration of a conventional susceptor;

[0059] FIG. **38** is a graph illustrating an example of an epi-crown height when an epitaxial layer is formed on a front surface of a semiconductor wafer using the susceptor illustrated in FIG. **37**;

[0060] FIG. **39** is an explanatory diagram illustrating a fluid simulation result regarding a fluid velocity of a source gas;

[0061] FIG. **40** is a graph illustrating an epi-crown height based on a difference in a method of mounting a SiC wafer; and

[0062] FIG. **41** is a graph illustrating an epi thickness based on a difference in a method of mounting a SiC wafer.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

Basis of Present Disclosure

[0063] As a basic technique related to a susceptor which is a wafer mounting apparatus of the present disclosure, a conventional example and problems thereof will be described. In recent years, there has been an increasing demand for single crystal epitaxial wafers containing SiC as a constituent material as a substrate for high withstand voltage electronic devices and the like.

[0064] When a power device is manufactured using a SiC single crystal wafer (hereinafter abbreviated as a “SiC wafer”), it is common to grow a SiC single crystal thin film on the SiC wafer by a thermal chemical vapor deposition method (hereinafter referred to as a “thermal chemical vapor deposition (CVD) method”), block a basal plane dislocation (BPD) on the SiC wafer, and form the device in a layer in which a doping concentration of impurities is controlled. When epitaxial growth is performed by a thermal CVD method, a SiC wafer is placed on a susceptor in an apparatus, and a silicon source gas such as a silane gas or a chlorosilane gas or a carbon source gas such as propane or methane is supplied together with a carrier gas such as hydrogen onto the wafer while the susceptor is rotated to perform epitaxial growth. The silicon source gas and the carbon source gas described above serve as source gases for epitaxial growth.

[0065] In order to prevent the SiC wafer from shifting during the epitaxial growth, the susceptor is generally provided with a counterbore (recess) corresponding to the wafer thickness, and the source gas described above is caused to laterally flow to the SiC wafer set therein so as to be substantially horizontal to the front surface of the SiC wafer.

[0066] In a SiC wafer of 6 inches or more, warping of the SiC wafer during epitaxial growth increases, and thus a bowl-shaped (recessed) counterbore bottom surface is often applied as a wafer mounting surface. By holding the SiC wafer at a wafer outer peripheral portion, it is possible to prevent contact between a wafer center portion and the counterbore bottom surface during film formation, and prevent a decrease in temperature of the wafer center portion. This effect improves in-plane uniformity of a film thickness and carrier concentration of an epitaxial layer.

[0067] The in-plane uniformity of the film thickness and the carrier concentration is an important index of the epitaxial growth layer. For example, film thickness unevenness is likely to occur in the outer peripheral portion of the SiC wafer, and a region of several mm from an outermost peripheral portion of the wafer cannot be used for device production. This unusable region is called edge exclusion or end cut, but if this region can be reduced, the region where semiconductor chips can be formed can be expanded, and the yield of the semiconductor chip can be improved. Therefore, it is preferable that the edge exclusion and end cut are small.

[0068] FIG. **29** is an explanatory view schematically illustrating an epi-crown phenomenon. As

illustrated in the drawing, an N.sup.- epitaxial layer **62** is formed on a front surface of a SiC wafer **61w**.

[0069] In epitaxial growth in the semiconductor field, as illustrated in FIG. **29**, it is known that the film thickness of the N.sup.- epitaxial layer **62** increases at the wafer outer peripheral portion, which is a region near an end portion on a front surface of the SiC wafer **61w**, and as a result, an epi-crown **62t** higher by an epi-crown height **H62** than a front surface of the other region of the N epitaxial layer **62** is generated. When the epi-crown **62t** is generated, the edge exclusion needs to be increased, so that the elimination of the epi-crown **62t** is required.

[0070] Several methods have been proposed so far for the resolution of epi-crown. The technique disclosed in Japanese Patent Application Laid-Open No. 7-226349 (1995) is a technique of making the film thickness of the wafer outer peripheral portion of the SiC wafer thinner than the other region in advance along with the growth of the epi-crown, and preventing the film thickness of the outer peripheral portion from becoming thicker than the other region even if the epi-crown occurs.

[0071] The technique disclosed in Japanese Patent Application Laid-Open No. 2014-27006 is a technique for removing the formed epi-crown after epitaxial growth. In addition, the technique disclosed in Japanese Patent Application Laid-Open No. 2016-119472 is a technique of increasing a wafer susceptor top distance, which is a step length between the front surface of the SiC wafer and the upper surface of the susceptor, in order to suppress supply of the source gas to the outer peripheral portion of the wafer.

[0072] However, in the method of thinning the outer peripheral portion of the wafer in advance as in the technology disclosed in Japanese Patent Application Laid-Open No. 7-226349 (1995), it is necessary to remove the outer peripheral portion in accordance with the film thickness of the epitaxial layer to be formed and the film thickness of the epi-crown to be formed at that time, and the wafer with a thinned outer peripheral portion is a dedicated wafer for forming an epitaxial layer having a specific thickness.

[0073] In an actual production site, since the specifications of the epitaxial layer required depending on the type of device to be manufactured are different, it is not desirable to prepare and stock wafers dedicated to specific film forming conditions since the production efficiency is significantly deteriorated.

[0074] As in the technique disclosed in Japanese Patent Application Laid-Open No. 2014-27006, in the case of removing the epi-crown after growing the epitaxial layer, since the vicinity of the removed epi-crown is scraped by polishing, there is also a problem that the epitaxial layer is roughened due to damage during polishing. A semiconductor device formed on such a polished surface is likely to be poor in performance, and thus the yield of the semiconductor chip is not necessarily improved. In addition, the methods disclosed in Japanese Patent Application Laid-Open No. 2014-27006 and Japanese Patent Application Laid-Open No. 7-226349 (1995) also have a problem that the number of steps for removing epi-crown increases.

[0075] On the other hand, the technique disclosed in Japanese Patent Application Laid-Open No. 2016-119472 is excellent in that there is no wafer processing as compared with Japanese Patent Application Laid-Open No. 2014-27006 and Japanese Patent Application Laid-Open No. 7-226349 (1995), but it has been newly found that there are the following problems as the examination proceeds.

[0076] FIGS. **30A** and **30B** are explanatory views schematically illustrating a conventional susceptor used for film formation. FIG. **30A** illustrates a planar configuration of the susceptor, and FIG. **30B** illustrates a cross-sectional structure of the susceptor.

[0077] As illustrated in FIGS. **30A** and **30B**, a SiC wafer **73** is accommodated in a circular counterbore **72** of a susceptor **71**. A source gas **G1** is supplied along a source supply direction **F1** above a front surface of the SiC wafer **73**. In this manner, the source gas **G1** serving as a source gas for epitaxial growth flowing along a flow **F2** of the source gas is supplied.

[0078] Here, as illustrated in FIG. **30B**, the distance between the uppermost surface of the

susceptor **71** and the front surface of the SiC wafer **73** is a wafer susceptor top distance DT. [0079] By executing a thermal CVD method of heating the SiC wafer **73** by a heating mechanism (not illustrated) provided on the susceptor **71** in a state where the source gas G1 is supplied, an epitaxial layer can be formed on the front surface of the SiC wafer **73**.

[0080] FIG. **31** is a graph illustrating a result of normalizing an epi-crown height (four points in a plane) when an epitaxial layer is grown on a front surface of a 6-inch SiC wafer **73** using the susceptor **71** illustrated in FIGS. **30A** and **30B** by an average epi thickness of the wafer. FIG. **32** is an explanatory diagram schematically illustrating a relationship between measurement points P1 to P4 and reference points B1 to B4 on the front surface of the SiC wafer **73**.

[0081] Four in-plane points illustrated in FIG. **31** are measurement points P1 to P4 along an outer periphery of the SiC wafer **73**. Note that an epi-crown height indicated by an epi-crown height change L1 in FIG. **31** indicates a difference between an epi thickness of the outermost peripheral portion and an epi thickness of a position at about 5 mm inward from the end of the SiC wafer **73**. That is, a difference in epi thickness between a measurement point Pi (any one of i=1 to 4) illustrated in FIG. **32** and a reference point Bi is illustrated. A distance DP illustrated in FIG. **32** indicates a distance of about 5 mm inward from the end of the SiC wafer **73**.

[0082] As illustrated in FIG. **31**, an epi-crown height at the measurement point P1 at which the measurement angle θ is 0° (360°) is the maximum value. As described above, it can be seen that a high epi-crown is formed protruding to the center of an orientation flat portion 73k in an outer peripheral portion of the front surface of the SiC wafer **73**. Hereinafter, the orientation flat center portion may be simply referred to as an “OF center portion”, and the orientation flat portion may be simply referred to as an “OF portion”.

[0083] An index corresponding to the “step between an upper surface of a susceptor and an upper surface of a silicon wafer” disclosed in Japanese Patent Application Laid-Open No. 2016-119472 is the wafer susceptor top distance DT illustrated in FIGS. **30A** and **30B**.

[0084] FIG. **33** is a graph illustrating a result of examining the correlation of the height of the epi-crown (normalized by an average epi thickness) at the OF center portion when the wafer susceptor top distance DT is changed. From an epi-crown height change L2 at the OF center portion illustrated in FIG. **33**, as the wafer susceptor top distance DT is increased, a result in which the epi-crown at the OF center portion is decreased is obtained, and the technique disclosed in Japanese Patent Application Laid-Open No. 2016-119472 seems to be effective. However, it has been newly found that the wafer susceptor top distance DT is a parameter that not only affects the epi-crown but also affects the epi thickness of the entire wafer outer peripheral portion.

[0085] FIG. **34** is a graph illustrating the correlation between the average epi thickness of the wafer outer peripheral portion and the wafer susceptor top distance DT. As indicated by measurement points P11 to P17 in FIG. **34**, the average epi thickness is a value obtained by normalizing measurement values at the seven measurement points P11 to P17 excluding the region located at 5 mm from the end portion of the SiC wafer **73** and close to the OF portion with the average epi thickness of the SiC wafer **73**.

[0086] From an average epi thickness change L3 in the wafer outer peripheral portion illustrated in FIG. **34**, it can be seen that the epi thickness in the outer peripheral portion of the SiC wafer **73** decreases as the wafer susceptor top distance DT increases.

[0087] FIG. **35** is a graph illustrating an epi thickness along an X direction when the wafer susceptor top distance DT is 0.7 mm. FIG. **36** is a graph illustrating an epi thickness along the X direction when the wafer susceptor top distance DT is 2.5 mm. Note that the epi thickness is indicated by a value normalized by the average epi thickness of the SiC wafer **73**.

[0088] In each of FIGS. **35** and **36**, the origin “0” intersecting the X axis and the Y axis is a position where a predetermined distance ΔD is “0”, the predetermined distance ΔD on a +X direction side is a positive distance [mm], and the predetermined distance ΔD on a -X direction side is a negative distance [mm].

[0089] From an epi thickness change LA illustrated in FIG. 35 and an epi thickness change L5 illustrated in FIG. 36, it can be seen well that the epi thickness of the outer peripheral portion of the SiC wafer 73 greatly changes as the wafer susceptor top distance DT increases. Furthermore, as can be seen from an epi thickness change L5 illustrated in FIG. 36, the epi thickness sharply decreases in the vicinity of the end portion of the SiC wafer 73 where the predetermined distance ΔD is about +70 mm.

[0090] Although the epi-crown generated in the outermost peripheral portion of the SiC wafer 73 can be suppressed, if the epi thickness of the wafer outer peripheral portion is significantly reduced, the technique disclosed in Japanese Patent Application Laid-Open No. 2016-119472 cannot be put into a semiconductor chip manufacturing process and reduces the wafer yield in the epitaxial growth process, which is not desirable.

[0091] A wafer mounting apparatus (susceptor) of the present disclosure is intended to reduce an ineffective region of a wafer by suppressing epi-crown without extremely reducing the epi thickness of the wafer outer peripheral portion.

[0092] The inventors of the wafer mounting apparatus of the present disclosure have conducted various studies to investigate a factor of forming a high epi-crown in the OF portion and a region in the vicinity thereof, and have found one factor. It is the distance between the wafer and the counterbore inner wall.

[0093] FIG. 37 is an explanatory view schematically illustrating a planar configuration of a conventional susceptor. As illustrated in the drawing, a susceptor 132 has three semiconductor wafers 131 such as SiC wafers mounted in a counterbore 133. The source gas G1 is supplied along the source supply direction F1 above the three semiconductor wafers 131 while rotating the susceptor 132 along a susceptor rotation direction R132.

[0094] FIG. 38 is a graph illustrating an example of an epi-crown height when an epitaxial layer is formed on a front surface of the semiconductor wafer 131 using the susceptor 132 illustrated in FIG. 37. Note that the epi-crown height indicated by an epi-crown height change L6 is normalized by the average epi thickness.

[0095] As illustrated in FIG. 38, the epi-crown has two peaks on the opposite side of the OF portion, and the portion indicating the peak corresponds to the portion indicated by "A" in FIG. 37. When considered together with the result illustrated in FIG. 31, it can be seen that the distance between the wafer outer periphery and the counterbore inner wall affects the epi-crown.

[0096] From these results, it can be said that it is important to use a counterbore whose counterbore inner wall and the wafer end are not extremely separated from each other, that is, a counterbore having a shape conforming to the wafer shape in order to suppress the epi-crown.

[0097] FIG. 39 is an explanatory diagram illustrating a fluid simulation result regarding the fluid velocity of the source gas G1. Dot concentration in the drawing has a negative correlation with a flow rate of the source gas G1, and a region having a higher dot concentration indicates that the flow rate of the source gas G1 is slower.

[0098] As illustrated in FIG. 39, the source gas G1 having passed over the susceptor 132 passes through the wafer mounting surface, collides with the end portion of the SiC wafer 73 such as the orientation flat portion 73k, and is then supplied to the front surface of the SiC wafer 73. That is, the end peripheral region R73 of the SiC wafer 73 is a region where the flow rate of the source gas G1 is the slowest.

[0099] Therefore, the end peripheral region R73 where the flow rate of the source gas G1 for epitaxial growth is the slowest is a region where the degree of epitaxial growth is the highest, and as a result, it is conceivable that epi-crown is formed in the end peripheral region R73.

[0100] Therefore, when the counterbore inner wall and the end portion of the SiC wafer 73 are close to each other, the source gas G1 is supplied to the front surface of the semiconductor wafer 131 without passing through the wafer mounting surface of the counterbore 133, and as a result, it is possible to avoid formation of a region where the flow velocity of the source gas G1 decreases,

such as the end peripheral region R73 in FIG. 39, and thus it is conceivable to be effective for suppressing epi-crown.

[0101] Next, a method of holding a semiconductor wafer was studied. FIG. 40 is a graph illustrating an epi-crown height based on a difference in a method of mounting a SiC wafer.

[0102] An epi-crown height change L8 indicates a normalized value of the epi thickness when a 6-inch SiC wafer is used as the SiC wafer 73 and the SiC wafer 73 is directly placed on the mounting surface. An epi-crown height change L7 indicates a normalized value of the epi thickness when the SiC wafer 73 is lifted up at several points and the wafer back surface and the susceptor are separated (hereinafter, this may be described as “lift-up”).

[0103] Note that the counterbore has a shape conforming to the shape of the SiC wafer 73, and the wafer susceptor top distance DT is equal between the epi-crown height change L7 and a epi-crown height change L8. As indicated by the epi-crown height change L8, in the case of direct placement even in the counterbore shape conforming to the shape of the SiC wafer 73, a high epi-crown is still formed in the OF portion. On the other hand, as indicated by the epi-crown height change L7, when the SiC wafer 73 is lifted up, the epi-crown of the OF portion decreases.

[0104] This is conceivably because a space is formed on the back surface side of the SiC wafer 73 by lift-up, and thus a part of the source gas G1 escapes to the back surface of the SiC wafer 73.

[0105] FIG. 41 is a graph illustrating an epi thickness along the X direction based on a difference in a method of mounting a SiC wafer. An epi thickness change L9 indicates a normalized value of the epi thickness when the SiC wafer 73 is lifted up, and an epi thickness change L10 indicates a normalized value of the epi thickness when the SiC wafer 73 is directly placed.

[0106] In FIG. 41, the origin “0” intersecting the X axis and the Y axis is a position where a predetermined distance ΔD is “0”, the predetermined distance ΔD on a +X direction side is a positive distance [mm], and the predetermined distance ΔD on a -X direction side is a negative distance [mm].

[0107] As indicated by the epi thickness changes L9 and L10 in FIG. 41, when both the wafer susceptor top distances DT are equal to each other, no significant change occurs in the in-plane distribution shape of the epi thickness.

[0108] From the above experimental results, the inventors have provided the wafer mounting apparatus of the present disclosure. In the wafer mounting apparatus of the present disclosure, a susceptor having an opening having a wall portion corresponding to the shape of the semiconductor wafer is provided so that the distance between a counterbore inner wall and a wafer end is not extremely separated, and the epitaxial growth process can be performed in a state where the semiconductor wafer is lifted up.

[0109] By lifting up and mounting the semiconductor wafer, a space is formed on the back surface of the semiconductor wafer, and a part of the source gas is released into the space, so that epi-crown can be reduced.

[0110] A point of epi-crown reduction is to release a part of the source gas G1 supplied to the surface of the semiconductor wafer to the back surface of the semiconductor wafer. Therefore, the wafer mounting apparatus of the present disclosure is characterized in that an annular groove having an annular shape is provided on the wafer mounting surface below the outermost peripheral portion of the semiconductor wafer to ensure a flow path of the source gas G1, and a flow path of the source gas G1 formed around the OF portion where the high epi-crown is formed is deeper than the other region.

[0111] The wafer mounting apparatus of the present disclosure obtained based on the above technical consideration is the wafer mounting apparatus described in the following first to fourth preferred embodiments.

Wafer Mounting Apparatus of Present Disclosure (Introduction)

[0112] Hereinafter, a susceptor which is a wafer mounting apparatus of the present disclosure and a

semiconductor manufacturing apparatus for a SiC wafer (SiC epitaxial wafer) will be described. Note that, in the drawings used in the following description, in order to facilitate understanding of characteristics of the wafer mounting apparatus of the present disclosure, characteristic portions may be illustrated in an enlarged manner for convenience, and the dimensional ratios and the like of the respective components may be different from the actual ones. In addition, materials, dimensions, and the like exemplified in the following description are merely examples, and the present disclosure is not limited thereto, and can be appropriately changed and implemented without changing the gist thereof.

(Semiconductor Manufacturing Apparatus)

[0113] FIG. 25 is an explanatory view schematically illustrating a cross-sectional structure of a semiconductor manufacturing apparatus 50 including a susceptor 11 and the like according to a first preferred embodiment of the present disclosure. The semiconductor manufacturing apparatus 50 is also called an “SiC epitaxial wafer manufacturing apparatus” and is an apparatus for forming an epitaxial layer on a front surface of a SiC wafer. Although the semiconductor manufacturing apparatus 50 is not limited to the configuration illustrated in FIG. 25, in order to facilitate understanding, a preferred embodiment described below will be described based on the semiconductor manufacturing apparatus 50 illustrated in FIG. 25.

[0114] The semiconductor manufacturing apparatus 50 illustrated in FIG. 25 is an apparatus called a planetary susceptor type, and includes a top plate 185 in an upper portion in a chamber 181, a planetary susceptor 182 and an induction heating coil 188 in a lower portion, and an injector 186 so as to vertically penetrate the chamber 181 from a center portion of the top plate 185.

[0115] In an upper portion of the planetary susceptor 182, a plurality of satellite disks 183 is arranged in a plurality of accommodation regions 189 around a center part 187, a SiC wafer 184 is set in the satellite disks 183, the source gas G1 radially flows from the injector 186, and epitaxial growth is performed on a front surface of the SiC wafer 184. Epitaxial growth on a front surface of the satellite disk 183 is performed in a temperature range of about 1400° C. to 1700° C. by a heating treatment using the induction heating coil 188.

[0116] Then, when the inside of the semiconductor manufacturing apparatus 50 reaches a desired temperature, a dopant gas is placed on the carrier gas in addition to a silicon-based gas (silicon source gas) and a hydrocarbon gas, the source gas G1 is made to flow from the side so as to be substantially horizontal to the SiC wafer 184, and an epitaxial growth process is performed. Silane (SiH.sub.4), dichlorosilane (SiCl.sub.2H.sub.2), trichlorosilane (SiCl.sub.3), silicon tetrachloride (SiCl.sub.4), and the like is conceivable as the silicon source gas, and propane (C.sub.3H.sub.8), ethane (C.sub.2H.sub.6), and the like is conceivable as the hydrocarbon gas as the carbon source gas.

[0117] For the purpose of improving the growth rate in the epitaxial growth process, HCl may be added or the source gas G1 containing Cl such as SiH.sub.2Cl.sub.2 may be used. Since the source gas G1 for epitaxial growth introduced from the injector 186 reaches the outside of the chamber 181 while being consumed for epitaxial growth, the growth rate and the carrier concentration are different between upstream and downstream. Thus, while the planetary susceptor 182 is mechanically rotated along a susceptor rotation direction R182 in a state where the source gas G1 is supplied, the satellite disk 183 is rotated along a disk rotation direction R183 by the disk rotation gas G2 blown onto a lower surface of the satellite disk 183.

[0118] Such an epitaxial growth process is executed to improve the uniformity of the carrier concentration and the film thickness in the epitaxial layer. When the epitaxial layer has a desired film thickness, the supply of the source gas G1 is stopped and the temperature is lowered. After falling to a predetermined temperature, the SiC wafer 184 is removed and a series of epitaxial growth processes ends.

[0119] Next, the susceptor of the present disclosure will be described. A ring is set in the satellite disk 183 described above to prevent displacement of the SiC wafer 184 during film formation. A

generic name of a wafer holder heated by induction heating by the induction heating coil **188**, a heater, or the like is referred to as a “susceptor”.

[0120] Therefore, in the preferred embodiment described below, there is no problem if a combination of a satellite disk and a ring is referred to as a susceptor, and hereinafter, for the sake of simplicity, a structure in which this combination is a main component is described as a “susceptor” of the present disclosure.

[0121] An advantage of the susceptor including two of the satellite disk and the ring is that the degree of freedom of usable materials and coating materials is increased, for example, by combining a SiC ring with a TaC-coated satellite disk. For example, when the epitaxial growth process is performed in an overlapping manner, it is conceivable that a deposit adhering to the inner wall of the ring interferes with the semiconductor wafer, and the semiconductor wafer cannot be set. However, when a SiC bulk ring is applied, SiC is strong, so that the deposit can be mechanically removed to enable reuse, and the part cost can be suppressed.

[0122] In addition, when the center portion of the semiconductor wafer is compared with an outer peripheral portion that is a peripheral region of the center portion, the temperature of the outer peripheral portion is likely to decrease. However, when a ring covered with some coating material is applied to a carbon base material and the resistivity of the base material of the ring is made higher than that of the base material of the satellite disk, the heat generation of the ring becomes larger than that of the satellite disk at the time of induction heating, and it is also possible to suppress the temperature decrease of the outer peripheral portion of the semiconductor wafer. In order to obtain the degree of freedom, the present disclosure employs the susceptor having the combination of a satellite disk and a ring as a main component.

First Preferred Embodiment

(Basic Configuration)

[0123] FIG. **1** is an explanatory view schematically illustrating a planar configuration of a susceptor **11** as a basic configuration of the first preferred embodiment of the present disclosure. FIG. **2** is an explanatory view schematically illustrating a configuration of an A1-A1 cross section of FIG. **1**, FIG. **3** is an explanatory view schematically illustrating a configuration of a B1-B1 cross section of FIG. **1**, and FIG. **4** is an explanatory view schematically illustrating a configuration of a C1-C1 cross section of FIG. **1**. FIG. **5** is an explanatory view illustrating an enlarged cross-sectional configuration of FIG. **2**. Note that, in FIG. **1**, planar structures of a ring **12** and a satellite disk **13**, which are main components of the susceptor **11**, are schematically illustrated in an easily recognizable manner.

[0124] A mounting target of the susceptor **11**, which is a basic configuration of the wafer mounting apparatus of the first preferred embodiment, is a SiC wafer **14** having an orientation flat portion **14k**. That is, the susceptor **11** of the first preferred embodiment is a wafer mounting apparatus on which the SiC wafer **14** having the orientation flat portion **14k** is mounted. The SiC wafer **14** is a semiconductor wafer having the orientation flat portion **14k** as a cutout portion.

[0125] As illustrated in FIGS. **1** to **5**, the susceptor **11** includes, as main components, a combination of the satellite disk **13**, which is a disk having a circular shape in plan view and a protruding cross section, and the ring **12**. The ring **12** is provided with a wall portion **12a** and a wall portion **12b** on an upper side, and has an opening O12 with the wall portion **12a** and the wall portion **12b** as outer peripheries. The opening O12 has a shape approximate to the wafer shape of the SiC wafer **14**, and the opening O12 has a planar shape including the SiC wafer **14** in plan view and slightly larger than the SiC wafer **14**.

[0126] As illustrated in FIGS. **2** to **5**, the cross-sectional structure of a portion having a terrace portion **12t** of the ring **12** has a deformed T-shaped cross-sectional structure having a region protruding in a direction approaching and a direction away from the opening O12.

[0127] The satellite disk **13**, which is a disk of the basic configuration, includes an upper layer portion **13u**, which is a protrusion, and includes a peripheral portion **13p** around the upper layer

portion **13u**. A peripheral portion front surface **13d** of the peripheral portion **13p** has a formation height lower than a front surface **13s** of the upper layer portion **13u**, and as illustrated in FIG. 5, a step **413** exists in a height direction from the peripheral portion surface **13d** to the front surface **13s**. [0128] The wall portion **12b** of the ring **12** functions as a cutout-side protruding wall portion protruding toward a side of the opening **O12** corresponding to the orientation flat portion **14k** which is a cutout portion of the SiC wafer **14**. As illustrated in FIG. 1, the wall portion **12b** is provided in parallel with the orientation flat portion **14k** in plan view, and a gap space **S12** in which a length in a lateral direction is a constant interval exists between the wall portion **12b** and the orientation flat portion **14k**.

[0129] As illustrated in FIGS. 2 and 5, an upper surface of the wall portion **12b** serving as the cutout-side protruding wall portion is provided to be a part of an upper surface of the entire ring **12**. That is, the formation height of the upper surface of the wall portion **12b** coincides with the formation height of the upper surface of the ring **12**. Furthermore, the lower surface of the wall portion **12b** does not have a contact relationship with the peripheral portion front surface **13d** of the peripheral portion **13p** of the satellite disk **13**.

[0130] The ring **12** further includes a terrace portion **12t** protruding toward the side of the opening **O12** in an intermediate region in the height direction, and as illustrated in FIG. 1, the terrace portion **12t** does not overlap the wall portion **12b**, which is the cutout-side protruding wall portion, in plan view.

[0131] The ring **12** is mounted on the satellite disk **13** in such a manner that the terrace portion **12t** of the ring **12** is supported by the peripheral portion front surface **13d** of the peripheral portion **13p** of the satellite disk **13**. That is, the satellite disk **13** supports the ring **12** by supporting the terrace portion **12t** from the back surface by the peripheral portion front surface **13d**.

[0132] The three lift-up pins **15**, which are a plurality of lift-up members each having the same formation height, are discretely mounted on the front surface of the terrace portion **12t** in the ring **12**.

[0133] The three lift-up pins **15** hold the SiC wafer **14** by supporting the SiC wafer **14** from the back surface. In the first preferred embodiment, the front surface of the terrace portion **12t** is a lift-up region for the three lift-up pins **15**.

[0134] Since the upper surface of the wall portion **12b** is a part of the upper surface of the ring **12**, the formation height of the upper surface of the wall portion **12b** coincides with the formation height of the upper surface of the entire ring **12**. The upper surface of the wall portion **12b** is set to be equal to or higher than the formation heights of the upper surfaces of three lift-up pins **15**.

[0135] As described above, the susceptor **11**, which is the basic configuration of the first preferred embodiment, includes the ring **12**, the satellite disk **13**, and the three lift-up pins **15** as main components.

[0136] In the susceptor **11** of the first preferred embodiment having such a configuration, the SiC wafer **14** can be held by mounting the SiC wafer **14** on the three lift-up pins **15** so that the SiC wafer **14** fits within the opening **O12** in plan view without being in contact with the ring **12**. This state is the semiconductor wafer mounting state related to the SiC wafer **14**.

[0137] In the semiconductor wafer state related to the SiC wafer **14**, the wall portion **12b**, which is the cutout-side protruding wall portion, and the orientation flat portion **14k** are in a positional relationship of facing each other via the gap space **S12** having a constant interval.

[0138] Since the SiC wafer **14** mounted in this manner is disposed in the opening **O12** in plan view without having a contact relationship with the ring **12**, an annular gap is always formed between an outer peripheral surface of the SiC wafer **14** and the opening **O12**. The annular gap includes the gap space **S12** described above.

[0139] In addition, since the opening **O12** is formed to approximate the shape of the SiC wafer **14**, the annular gap is provided at a constant interval without bias. This annular gap also functions as a gas escape flow path to be described later.

[0140] When the SiC wafer **14** is mounted on the three lift-up pins **15**, the disk upper space **S13** is formed between the front surface **13s** of the upper layer portion **13u** of the satellite disk **13** and the back surface of the SiC wafer **14** except for a formation region of the three lift-up pins **15**.

[0141] Furthermore, a groove **2**, which is an annular groove having an annular shape along the outer periphery of the opening **O12** in plan view, is formed between the upper layer portion **13u** of the satellite disk **13** and the ring **12** on the back surface side of the SiC wafer **14**.

[0142] The annular groove **2** has one side surface as a side surface of the upper layer portion **13u** and the other side surface as the wall portion **12a**. Then, the groove **2** formed in the formation region of the terrace portion **12t** is a space above the terrace portion **12t** with the terrace portion **12t** as a bottom surface.

[0143] On the other hand, as illustrated in FIGS. **2** and **5**, in the region where the terrace portion **12t** is not formed, the groove **2** immediately below the wall portion **12b** is a groove **2k**. Further, as illustrated in FIG. **4**, the groove **2** located immediately below both end peripheral regions of the wall portion **12b** is a groove **2m**. Both the groove **2k** and the groove **2m** are spaces above the peripheral portion front surface **13d** with the peripheral portion front surface **13d** of the peripheral portion **13p** as a bottom surface.

[0144] Hereinafter, in the present specification, the annular groove **2** will be described as including the groove **2k** (groove **2n**) and the groove **2m** (groove **2m2**) in a broad sense, and indicating only the groove **2** with the terrace portion **12t** as a bottom surface in a narrow sense.

[0145] The annular gap formed between the outer peripheral surface of the SiC wafer **14** and the opening **O12** and the groove **2** which is an annular groove described above serve as a gas escape flow path for releasing the source gas **G1** supplied to the front surface of the SiC wafer **14** to the disk upper space **S13**.

[0146] The terrace portion **12t** of the ring **12** is provided so as not to have a contact relationship with the side surface of the upper layer portion **13u** of the satellite disk **13**. Therefore, an auxiliary groove **2X** communicating with the groove **2** in the depth direction can be provided between a front end of the terrace portion **12t** and the side surface of the upper layer portion **13u**. The auxiliary groove **2X** makes it possible to increase the formation depth of a part of the groove **2** and to increase the volume of the gas escape flow path.

[0147] In FIGS. **1**, **2**, and **5**, the lift-up pins **15** are illustrated in a cylindrical shape, but may have any shape, and while at least three lift-up pins are necessary for holding the SiC wafer **14**, the number of lift-up pins is not limited as long as three or more lift-up pins are provided.

[0148] A place where the lift-up pin **15** and the SiC wafer **14** are in contact with each other has a temperature different from other regions and may affect film quality, and thus, for example, it is desirable to hold the SiC wafer **14** in an edge exclusion region. Therefore, it is preferable that the position of the lift-up pin **15** is provided on the outside as much as possible so as to be in contact with the wall portion **12a** of the ring **12**.

[0149] There is no problem even if the lift-up pin **15** is provided so as not to come into contact with the wall portion **12a**, but there is a possibility that it becomes difficult to manufacture the ring **12** by machining or the like with an NC machining machine.

[0150] As illustrated in FIGS. **2** to **5**, the terrace portion **12t** is formed to have a thickness in the height direction smaller than the step difference $\Delta 13$ between the upper layer portion **13u** and the peripheral portion **13p** in the satellite disk **13**. Therefore, when the satellite disk **13** and the ring **12** are combined with each other, the annular groove **2** is always formed also in the formation region of the terrace portion **12t**.

[0151] As described above, the formation position of the lower surface of the wall portion **12b** is located between the front surface of the ring **12** and the back surface of the SiC wafer **14**, but is desirably located between the front surface and the back surface of the SiC wafer **14**.

[0152] In FIG. **5**, the lower surface of the wall portion **12b** corresponding to the orientation flat portion **14k** of the SiC wafer **14** is aligned in height with the upper surfaces of the three lift-up pins

15 (the back surface of the SiC wafer **14**). In addition, the ring **12** is configured such that the terrace portion **12t** is not formed at a location corresponding to the orientation flat portion **14k**. Therefore, the groove **2k** serving as a gas escape flow path for the source gas **G1** to escape into the disk upper space **S13** on the back surface side of the SiC wafer **14** can be formed deeper than the groove **2** with the terrace portion **12t** as a bottom surface.

[0153] Similarly, as illustrated in FIG. **4**, since the terrace portion **12t** is not formed in the end peripheral region of the orientation flat portion **14k**, the groove **2m** serving as the gas escape flow path can be made deeper than the groove **2** with the terrace portion **12t** as a bottom surface.

[0154] When the susceptor **11** of the first preferred embodiment having such a configuration is applied to the semiconductor manufacturing apparatus **50** illustrated in FIG. **25**, the satellite disk **183** is replaced with the susceptor **11** including the ring **12**, the satellite disk **13**, and the three lift-up pins **15**, and the SiC wafer **184** is replaced with the SiC wafer **14**.

[0155] When the susceptor **11** of the first preferred embodiment illustrated in FIGS. **1** to **5** is applied to the semiconductor manufacturing apparatus **50** illustrated in FIG. **25**, the source gas **G1** is supplied from the outside of the rotating susceptor **132** in the planetary susceptor type CVD apparatus. When the orientation flat portion **14k** of the SiC wafer **14** approaches the injector **186** and the source gas **G1** is supplied from the orientation flat portion **14k** side, a part of the source gas **G1** flows into the disk upper space **S13** through the gap space **S12** and the groove **2k** between the SiC wafer **14** and the wall portion **12b**.

[0156] As described above, since the groove **2k** is formed relatively deep, the source gas **G1** flowing from the gap space **S12** between the SiC wafer **14** and the wall portion **12b** can be smoothly released to the disk upper space **S13** on the back surface side of the SiC wafer **14** via the groove **2k** serving as the gas escape flow path.

[0157] On the other hand, when the lower surface of the wall portion **12b** is lower than the back surface of the SiC wafer **14**, the wall portion **12b** protrudes to the groove **2k**, and there is a concern that the function of the groove **2k** as a gas escape flow path is hindered. Therefore, the position of the lower surface of the wall portion **12b** is set to be equal to or higher than the height of the front surfaces of the three lift-up pins **15** (the back surface of the SiC wafer **14**).

[0158] In the susceptor **11**, which is the basic configuration of the first preferred embodiment, the terrace portion **12t** is supported from below on the front surface of the peripheral portion **13p** of the satellite disk **13** to hold the ring **12**, and the formation of the terrace portion **12t** in the orientation flat portion **14k** and the region in the vicinity thereof is omitted.

[0159] However, if there is no problem in terms of strength, a plurality of partial terrace portions may be discretely provided instead of the terrace portion **12t** continuously formed in a region other than the orientation flat portion **14k** and the neighboring region thereof. Note that, here, "there is no problem in terms of strength" means that the weight of the ring **12** can be supported by the terrace portion **12t**, and no problem occurs in the terrace portion **12t**.

[0160] In the susceptor **11**, which is the basic configuration of the wafer mounting apparatus according to the first preferred embodiment of the present disclosure, the SiC wafer **14**, which is a semiconductor wafer, is mounted on the three lift-up pins **15**, which are a plurality of lift-up members, to achieve the semiconductor wafer mounting state related to the SiC wafer **14**.

Therefore, the disk upper space **S13** can be provided between the back surface of the SiC wafer **14** and the front surface **13s** of the satellite disk **13**.

[0161] Further, in the susceptor **11** of the first preferred embodiment, the groove **2** which is an annular groove is formed between the upper layer portion **13u** of the satellite disk **13** and the ring **12** on the back surface side of the SiC wafer **14** during the semiconductor wafer mounting state.

[0162] Therefore, when the source gas **G1** is supplied above the front surface of the SiC wafer **14** in the semiconductor wafer mounting state, a part of the source gas **G1** can be intentionally released to the disk upper space **S13** on the back surface side of the SiC wafer **14** through the annular gap and the groove **2** along the outer periphery of the SiC wafer **14**.

[0163] As a result, when the susceptor **11** of the first preferred embodiment executes the epitaxial growth process to form an epitaxial growth layer on the front surface of the SiC wafer **14** in the semiconductor wafer mounting state, the susceptor can suppress the occurrence of an epi-crown phenomenon in which the epi thickness of the epitaxial growth layer locally increases. This epi-crown phenomenon occurs mainly on the outer peripheral portion on the front surface of the SiC wafer **14**.

[0164] As a result, the susceptor **11** of the first preferred embodiment has an effect of being able to form an epitaxial layer having an epi thickness with excellent in-plane uniformity on the front surface of the SiC wafer **14** in the semiconductor wafer mounting state. Note that an epi thickness and a carrier concentration (impurity concentration) are included as indices of in-plane uniformity.

[0165] Furthermore, since the height of the uppermost surface of the ring **12** can be arbitrarily set, the wafer susceptor top distance DT, which is the difference in height between the front surface of the SiC wafer **14** mounted on the three lift-up pins **15** and the uppermost surface of the ring **12**, can be set to an optimum length that does not cause an extreme decrease in the epi thickness at the outer peripheral portion of the front surface of the SiC wafer **14**.

[0166] In the susceptor **11** of the first preferred embodiment, since the ring **12** is mounted on the satellite disk **13** in a mode in which the terrace portion **12t** is supported on the peripheral portion front surface **13d** of the peripheral portion **13p** of the satellite disk **13**, a combination structure of the satellite disk **13** and the ring **12** can be achieved without affecting the SiC wafer **14** as a mounting target.

[0167] Furthermore, by forming the thickness of the terrace portion **12t** in the height direction to be thinner than the step $\Delta 13$ between the upper layer portion **13u** and the peripheral portion **13p** of the satellite disk **13**, the groove **2** with the terrace portion **12t** as a bottom surface can be formed in the formation region of the terrace portion **12t**.

[0168] The susceptor **11** of the first preferred embodiment can support the SiC wafer **14** by the three lift-up pins **15**, which are a plurality of lift-up members mounted on the terrace portion **12t** of the ring **12**, with the lift-up region as the surface region of the terrace portion **12t**.

[0169] In the susceptor **11** of the first preferred embodiment, since the terrace portion **12t** does not have a contact relationship with the side surface of the upper layer portion **13u**, the auxiliary groove **2X** communicating with the groove **2** in the depth direction can be formed between the side surface of the upper layer portion **13u** and the protruding distal end portion of the terrace portion **12t**.

[0170] As a result, since the susceptor **11** of the first preferred embodiment can increase the volume of the groove **2** by the amount of the auxiliary groove **2X**, it is possible to improve the function as a gas escape flow path by the groove **2** to release a part of the source gas G1 to the disk upper space S13 on the back surface side of the SiC wafer **14**, and to effectively suppress the epi-crown phenomenon.

[0171] Since the opening O12 of the ring **12** is shaped to include the SiC wafer **14** and has a larger formation area than the SiC wafer **14**, the SiC wafer **14** can be reliably mounted on the three lift-up pins **15** without being brought into contact with the ring **12** including the wall portion **12b** serving as the cutout-side protruding wall portion.

[0172] Furthermore, by making the opening O12 have an approximate shape conforming to the shape of the SiC wafer **14**, the susceptor **11** of the first preferred embodiment can effectively suppress the occurrence of epi-crown.

[0173] In addition, since the wall portion **12b**, which is the cutout-side protruding wall portion, does not have a contact relationship with the peripheral portion front surface **13d** of the peripheral portion **13p** of the satellite disk **13**, the function as the gas escape flow path of the groove **2k** formed below the wall portion **12b** is not deteriorated.

[0174] In the susceptor **11** of the first preferred embodiment, since the formation height of the lower surface of the wall portion **12b**, which is the cutout-side protruding wall portion, is set to be

equal to or higher than the formation heights of the upper surfaces of the three lift-up pins **15**, the function of the groove **2k** formed below the wall portion **12b** as the gas escape flow path is not deteriorated.

[0175] In addition, in the susceptor **11** of the first preferred embodiment, the upper surface of the wall portion **12b**, which is the cutout-side protruding wall portion, is provided to be a part of the upper surface of the ring **12**. That is, the formation height of the upper surface of the wall portion **12b** coincides with the formation height of the upper surface of the ring **12**. Therefore, the groove **2k** as the gas escape flow path can be formed below the wall portion **12b** having a sufficient thickness.

[0176] Since the wall portion **12b**, which is the cutout-side protruding wall portion, is disposed to face the orientation flat portion **14k** via the gap space **S12** having a constant interval during the semiconductor wafer mounting state, the amount of the source gas **G1** flowing through the groove **2k** via the gap space **S12** can be limited so as not to exceed an allowable upper limit amount. This is because if the source gas **G1** exceeding the allowable upper limit amount flows into the groove **2k** through the gap space **S12**, the source gas **G1** supplied to the front surface of the SiC wafer **14** may fall below an appropriate amount.

[0177] Therefore, the susceptor **11** of the first preferred embodiment can supply an appropriate amount of the source gas **G1** to the front surface of the SiC wafer **14** having the orientation flat portion **14k**.

First Modification

[0178] FIG. **6** is an explanatory view schematically illustrating a planar configuration of a susceptor **11B** according to a first modification of the first preferred embodiment of the present disclosure. FIG. **7** is an explanatory view schematically illustrating a configuration of a cross section taken along line **A11-A11** in FIG. **6**, FIG. **8** is an explanatory view schematically illustrating a configuration of a cross section taken along line **B11-B11** in FIG. **6**, and FIG. **9** is an explanatory view schematically illustrating a configuration of a cross section taken along line **C11-C11** in FIG. **6**. Note that, in FIG. **6**, the planar structures of the ring **12** and a satellite disk **13B**, which are main components of the susceptor **11B**, are schematically illustrated in an easily recognizable manner.

[0179] Similarly to the basic configuration, a mounting target of the susceptor **11B** according to the first modification of the wafer mounting apparatus of the first preferred embodiment is the SiC wafer **14** having the orientation flat portion **14k**.

[0180] Hereinafter, components similar to those of the susceptor **11** having the basic configuration illustrated in FIGS. **1** to **5** will be denoted by the same reference numerals, description thereof will be omitted as appropriate, and features of the susceptor **11B** as the first modification will be mainly described.

[0181] As illustrated in FIGS. **6** to **9**, the susceptor **11B** includes, as main components, a combination of a satellite disk **13B**, which is a disk having a circular shape in plan view and a protruding cross section, and a ring **12** having a deformed T-shaped cross-sectional structure having a terrace portion **12t**.

[0182] The satellite disk **13B** that is the disk of the first modification includes an upper layer portion **13u** that is a protrusion, and a peripheral portion **13p** around the upper layer portion **13u**.

[0183] The front surface **13s** of the upper layer portion **13u** in the satellite disk **13B** has a concave shape in which the center portion is recessed. That is, the front surface **13s** has a concave portion that becomes deeper from the outer peripheral portion toward the center portion.

[0184] In the susceptor **11B** according to the first modification of the first preferred embodiment having such a configuration, similarly to the basic configuration, when the SiC wafer **14** is mounted on the three lift-up pins **15**, a disk upper space **S13B** is formed between the front surface **13s** of the upper layer portion **13u** of the satellite disk **13B** and the back surface of the SiC wafer **14** except for the formation region of the three lift-up pins **15**.

[0185] The disk upper space **S13B** has a larger volume than the disk upper space **S13** of the basic configuration as the formation depth increases from the outer periphery to the center portion of the SiC wafer **14** on the back surface side of the SiC wafer **14** as compared with the disk upper space **S13**.

[0186] The susceptor **11B** of the first modification has effects similar to those of the susceptor **11** of the basic configuration, and further has the following unique effects.

[0187] Since the front surface **13s** of the upper layer portion **13u** of the satellite disk **13** in the susceptor **11B** of the first modification has a concave shape in which the center portion is recessed, it is possible to form the disk upper space **S13B** larger than the disk upper space **S13** of the basic configuration on the back surface side of the SiC wafer **14**.

[0188] As a result, the susceptor **11B** of the first modification can release a part of the source gas **G1** to the relatively large disk upper space **S13B** via the groove **2** which is an annular groove, so that the flow of the source gas **G1** to be released to the back surface side of the SiC wafer **14** can be made smoother.

Second Modification

[0189] FIG. **10** is an explanatory view schematically illustrating a planar configuration of a susceptor **11C** according to a second modification of the first preferred embodiment of the present disclosure. FIG. **11** is an explanatory view schematically illustrating a configuration of a cross section taken along line **A12-A12** of FIG. **10**, FIG. **12** is an explanatory view schematically illustrating a configuration of a cross section taken along line **B12-B12** of FIG. **10**, and FIG. **13** is an explanatory view schematically illustrating a configuration of a cross section taken along line **C12-C12** of FIG. **10**. Note that, in FIG. **10**, planar structures of a ring **121** and a satellite disk **13**, which are main components of the susceptor **11C**, are schematically illustrated in an easily recognizable manner.

[0190] Similarly to the basic configuration, a mounting target of the susceptor **11C**, which is the second modification of the wafer mounting apparatus of the first preferred embodiment, is the SiC wafer **14** having the orientation flat portion **14k**.

[0191] Hereinafter, components similar to those of the susceptor **11** of the basic configuration illustrated in FIGS. **1** to **5** will be denoted by the same reference numerals, description thereof will be omitted as appropriate, and features of the susceptor **11C** as the second modification will be mainly described.

[0192] As illustrated in FIGS. **10** to **13**, the susceptor **11C** includes, as main components, a combination of a satellite disk **13**, which is a disk having a circular shape in plan view and a protruding cross section, and a ring **121** having a deformed T-shaped cross-sectional structure having a terrace portion **121t**.

[0193] The ring **121** in the susceptor **11C** of the second modification is characterized by further including two terrace portions **121t**, which are auxiliary terrace portions protruding toward the side of the opening **O12**, in addition to the terrace portion **12t**.

[0194] As illustrated in FIG. **10**, the two terrace portions **121t**, which are auxiliary terrace portions, are provided adjacent to both ends of the wall portion **12b**. That is, the two terrace portions **121t** are provided between the terrace portion **12t** and the wall portion **12b** in plan view.

[0195] As illustrated in FIG. **13**, formation heights of back surfaces of the two terrace portions **121t** coincide with the formation height of the back surface of the terrace portion **12t**. Then, the terrace portion **12t** has a terrace thickness **T1** in the height direction, and the terrace portions **121t** have a terrace thickness **T2** ($<T1$) in the height direction. As described above, the terrace portions **121t**, which are auxiliary terrace portions, are set to be thinner in the height direction than the terrace portion **12t**.

[0196] As illustrated in FIG. **13**, the ring **121** is mounted on the satellite disk **13** in a mode in which the terrace portion **12t** and the terrace portions **121t** are supported on the peripheral portion front surface **13d** of the peripheral portion **13p** of the satellite disk **13**. Spaces above the terrace portions

121t are grooves **2m2** serving as a gas escape flow path.

[0197] That is, in the groove **2m**, the terrace portion **121t** is a bottom surface, the side surface of the upper layer portion **13u** is one side surface, and the wall portion **12a** is the other side surface.

[0198] The susceptor **11C** of the second modification having such a configuration has effects similar to those of the susceptor **11** of the basic configuration, and further has the following unique effects.

[0199] In the susceptor **11C**, which is the wafer mounting apparatus of the second modification, the ring **121** is mounted on the satellite disk **13** in a mode in which the terrace portion **12t** and the terrace portions **121t** are supported on the peripheral portion front surface **13d** of the peripheral portion **13p** of the satellite disk **13**, so that the combination structure of the satellite disk **13** and the ring **121** can be achieved with good stability.

[0200] Furthermore, by setting the terrace portions **121t**, which are auxiliary terrace portions, to be thinner in the height direction than the terrace portion **12t**, the grooves **2m2**, which are spaces above the terrace portions **121t**, can be formed deeper than the groove **2**, which is a space above the terrace portion **12t**. That is, formation depths of the grooves **2m2**, which are peripheral regions of both ends of the orientation flat portion **14k**, can be made deeper than the formation depth of the groove **2** with the terrace portion **12t** as a bottom surface.

[0201] As a result, the susceptor **11C** of the second modification can minimize deterioration of the function of the groove **2m2** as a gas escape flow path for releasing the source gas **G1** supplied to the periphery of the orientation flat portion **14k** as a cutout portion to the disk upper space **S13** on the back surface side of the SiC wafer **14**.

Second Preferred Embodiment

(Basic Configuration)

[0202] FIG. **14** is an explanatory view schematically illustrating a planar configuration of a susceptor **21**, which is a basic configuration of a second preferred embodiment of the present disclosure. FIG. **15** is an explanatory view schematically illustrating a configuration of an A2-A2 cross section of FIG. **14**, FIG. **16** is an explanatory view schematically illustrating a configuration of a B2-B2 cross section of FIG. **14**, and FIG. **17** is an enlarged explanatory view illustrating a focused region **R2** of FIG. **14**. Note that, in FIG. **14**, the planar structures of the ring **122** and the satellite disk **13**, which are main components of the susceptor **21**, are schematically illustrated in an easily recognizable manner.

[0203] A mounting target of the susceptor **21**, which is the basic configuration of the wafer mounting apparatus according to the second preferred embodiment, is a SiC wafer **24** having a notch **24n** having a notch length L_n (see FIG. **17**). That is, the susceptor **21** of the second preferred embodiment is a wafer mounting apparatus on which the SiC wafer **24** having the notch **24n** is mounted. The SiC wafer **24** is a semiconductor wafer having the notch **24n** as a cutout portion.

[0204] Hereinafter, components similar to those of the susceptor **11** of the basic configuration of the first preferred embodiment illustrated in FIGS. **1** to **5** will be denoted by the same reference numerals, description thereof will be omitted as appropriate, and features of the susceptor **21** of the basic configuration of the second preferred embodiment will be mainly described.

[0205] As illustrated in FIGS. **14** to **17**, the susceptor **21** includes, as main components, a combination of a satellite disk **13**, which is a disk having a circular shape in plan view and a protruding cross section, and a ring **122** having a deformed T-shaped cross-sectional structure having a terrace portion **12t**.

[0206] The ring **122** is provided with a wall portion **12a** and a wall portion **12c** on an upper side, and has an opening **O122** with the wall portion **12a** and the wall portion **12c** as outer peripheries. The opening **O122** has a shape approximate to the wafer shape of the SiC wafer **24**, includes the SiC wafer **24** in plan view, and has a planar shape slightly larger than the SiC wafer **24**.

[0207] The wall portion **12c** of the ring **122** functions as a cutout-side protruding wall portion protruding toward a side of the opening **O122** corresponding to the notch **24n**, which is a cutout

portion of the SiC wafer **24**. As illustrated in FIG. **14**, the wall portion **12c**, which is the cutout-side protruding wall portion, is disposed so as to fill a part of the notch **24n** without coming into contact with the SiC wafer **24**, which is a semiconductor wafer.

[0208] As illustrated in FIG. **15**, the wall portion **12c** serving as the cutout-side protruding wall portion does not have a contact relationship with the peripheral portion front surface **13d** of the peripheral portion **13p** of the satellite disk **13**.

[0209] The ring **122** further includes a terrace portion **12t** protruding toward the side of the opening **O122** in an intermediate region in the height direction, and as illustrated in FIG. **14**, the terrace portion **12t** does not overlap the wall portion **12c**, which is the cutout-side protruding wall portion, in plan view.

[0210] The ring **122** is mounted on the satellite disk **13** in such a manner that the terrace portion **12t** is supported on the peripheral portion front surface **13d** of the peripheral portion **13p** of the satellite disk **13**.

[0211] In the susceptor **21**, which is the basic configuration of the second preferred embodiment, a front surface of the terrace portion **12t** is a lift-up region for the three lift-up pins **15**.

[0212] The three lift-up pins **15**, which are a plurality of lift-up members each having the same height, are mounted on the front surface of the terrace portion **12t** in the ring **122**. The three lift-up pins **15** hold the SiC wafer **24** by supporting the SiC wafer **24** from a back surface.

[0213] As described above, the susceptor **21**, which is the basic configuration of the second preferred embodiment, includes the ring **121**, the satellite disk **13**, and the three lift-up pins **15** as main components.

[0214] In the susceptor **21** of the second preferred embodiment having such a configuration, the SiC wafer **24** can be held by mounting the SiC wafer **24** on the three lift-up pins **15** so that the SiC wafer **24** fits within the opening **O122** in plan view without being in contact with the ring **122**. This state is the semiconductor wafer mounting state related to the SiC wafer **24**.

[0215] During the semiconductor wafer mounting state related to the SiC wafer **24**, the wall portion **12c**, which is the cutout-side protruding wall portion, faces the notch **24n** and is arranged so as to fill a part of the notch **24n**.

[0216] Since the SiC wafer **24** mounted in this manner is disposed in the opening **O122** in plan view without having a contact relationship with the ring **122**, an annular gap is always formed between an outer peripheral surface of the SiC wafer **24** and the opening **O122**. In addition, since the opening **O122** is formed to approximate the shape of the SiC wafer **24**, the annular gaps are provided at a constant interval without bias. This annular gap also functions as a gas escape flow path.

[0217] When the SiC wafer **24** is mounted on the three lift-up pins **15**, a disk upper space **S13** is formed between the front surface **13s** of the upper layer portion **13u** of the satellite disk **13** and the back surface of the SiC wafer **24** except for a formation region of the three lift-up pins **15**.

[0218] Furthermore, a groove **2**, which is an annular groove having an annular shape along the outer periphery of the opening **O122** in plan view, is formed between the upper layer portion **13u** of the satellite disk **13** and the ring **122** on the back surface side of the SiC wafer **24**.

[0219] The annular groove **2** has one side surface as a side surface of the upper layer portion **13u** and the other side surface as the wall portion **12a**. Then, the groove **2** formed in the formation region of the terrace portion **12t** is a space above the terrace portion **12t** with the terrace portion **12t** as a bottom surface.

[0220] On the other hand, as illustrated in FIG. **15**, in the region where the terrace portion **12t** is not formed, the groove **2** immediately below the wall portion **12c** is a groove **2n**. The groove **2n** is a space above the peripheral portion front surface **13d** with the peripheral portion front surface **13d** of the peripheral portion **13p** as a bottom surface.

[0221] The terrace portion **12t** of the ring **122** is provided so as not to have a contact relationship with the side surface of the upper layer portion **13u** of the satellite disk **13**. Therefore, an auxiliary

groove **2X** communicating with the groove **2** in the depth direction can be provided between the front end of the terrace portion **12t** and the side surface of the upper layer portion **13u**.

[0222] The lower surface of the wall portion **12c** is positioned between the front surface of the ring **122** and the back surface of the SiC wafer **24** similarly to the wall portion **12b** of the preferred embodiment, but is desirably positioned between the front surface and the back surface of the SiC wafer **24**.

[0223] In the susceptor **21** of the second preferred embodiment, the lower surface of the wall portion **12c** corresponding to the notch **24n** of the SiC wafer **24** is aligned in height with the upper surfaces of the three lift-up pins **15** (the back surface of the SiC wafer **24**). In addition, the ring **122** is configured so that the terrace portion **12t** is not formed at a location corresponding to the notch **24n** and a region in the vicinity thereof. Therefore, the groove **2n** serving as a gas escape flow path for releasing the source gas **G1** to the disk upper space **S13** can be provided deeper than the groove **2** with the terrace portion **12t** as a bottom surface.

[0224] When the susceptor **21** of the second preferred embodiment is applied to the semiconductor manufacturing apparatus **50** illustrated in FIG. **25**, the satellite disk **183** is replaced with the susceptor **21** including the ring **122**, the satellite disk **13**, and the three lift-up pins **15**, and the SiC wafer **184** is replaced with the SiC wafer **24**.

[0225] In a case where the susceptor **21** of the second preferred embodiment illustrated in FIGS. **14** to **17** is applied to the semiconductor manufacturing apparatus **50** illustrated in FIG. **25**, the source gas **G1** is supplied from the outside of the rotating susceptor **132** in the planetary susceptor type CVD apparatus. When the notch **24n** of the SiC wafer **24** approaches the injector **186** and the source gas **G1** is supplied from the notch **24n** side, a part of the source gas **G1** flows into the disk upper space **S13** through the gap space between the SiC wafer **24** and the wall portion **12c** and the groove **2n**.

[0226] As described above, since the groove **2n** is formed deeper than the groove **2** with the terrace portion **12t** as a bottom surface, the source gas **G1** supplied above the notch **24n** of the SiC wafer **24** can be smoothly released to the disk upper space **S13** on the back surface side of the SiC wafer **24**.

[0227] On the other hand, when the lower surface of the wall portion **12c** is lower than the back surface of the SiC wafer **24**, the wall portion **12c** protrudes to the groove **2n** and there is a concern that the function of the groove **2n** as a gas escape flow path may be impaired, and thus the position of the lower surface of the wall portion **12c** is set to be equal to or higher than the heights of the upper surfaces of the three lift-up pins **15** (back surface of the SiC wafer **24**).

[0228] The susceptor **21**, which is the basic configuration of the wafer mounting apparatus according to the second preferred embodiment of the present disclosure, achieves the semiconductor wafer mounting state related to the SiC wafer **24** by mounting the SiC wafer **24**, which is a semiconductor wafer, on the three lift-up pins **15**. Therefore, the disk upper space **S13** can be provided between the back surface of the SiC wafer **24** and the front surface **13s** of the satellite disk **13**.

[0229] Furthermore, in the susceptor **21** of the second preferred embodiment, the groove **2** which is an annular groove is formed between the upper layer portion **13u** of the satellite disk **13** and the ring **122** on the back surface side of the SiC wafer **24** during the semiconductor wafer mounting state.

[0230] Therefore, when the source gas **G1** is supplied above the front surface of the SiC wafer **24** in the semiconductor wafer mounting state, a part of the source gas **G1** can be intentionally released to the disk upper space **S13** on the back surface side of the SiC wafer **24** through the annular gap and the groove **2** along the outer periphery of the SiC wafer **24**.

[0231] As a result, when the susceptor **21** according to the second preferred embodiment executes the epitaxial growth process to form an epitaxial growth layer on the front surface of the SiC wafer **24** in the semiconductor wafer mounting state, the susceptor can suppress the occurrence of an epi-

crown phenomenon in which the epi thickness of the epitaxial growth layer locally increases. This epi-crown phenomenon occurs mainly on the outer peripheral portion on the front surface of the SiC wafer **24**.

[0232] As a result, the susceptor **21** according to the second preferred embodiment has an effect of being able to form an epitaxial layer having excellent in-plane uniformity on the front surface of the SiC wafer **24** in the semiconductor wafer mounting state. Note that an epi thickness and a carrier concentration (impurity concentration) are included as indices of in-plane uniformity.

[0233] Furthermore, since the height of the uppermost surface of the ring **122** can be arbitrarily set, the wafer susceptor top distance DT, which is the difference in height between the front surface of the SiC wafer **24** mounted on the three lift-up pins **15** and the uppermost surface of the ring **122**, can be set to an optimum length that does not cause an extreme decrease in the epi thickness at the outer peripheral portion of the front surface of the SiC wafer **24**.

[0234] Since the opening O**122** of the ring **122** is shaped to include the SiC wafer **24** and has a larger formation area than the SiC wafer **24**, the SiC wafer **24** can be reliably mounted on the three lift-up pins **15** without being brought into contact with the ring **122** including the wall portion **12c** serving as the cutout-side protruding wall portion.

[0235] Furthermore, by making the opening O**122** have an approximate shape conforming to the shape of the SiC wafer **24**, the susceptor **21** according to the second preferred embodiment can effectively suppress the occurrence of epi-crown.

[0236] In addition, since the wall portion **12c**, which is the cutout-side protruding wall portion, does not have a contact relationship with the peripheral portion front surface **13d** of the peripheral portion **13p** of the satellite disk **13**, the function as the gas escape flow path of the groove **2n** formed below the wall portion **12c** is not deteriorated.

[0237] In the susceptor **21** of the second preferred embodiment, since the formation height of the lower surface of the wall portion **12c**, which is the cutout-side protruding wall portion, is set to be equal to or higher than the formation heights of the upper surfaces of the three lift-up pins **15**, the function of the groove **2n** as the gas escape flow path is not deteriorated.

[0238] In addition, in the susceptor **21** of the second preferred embodiment, the upper surface of the wall portion **12c**, which is the cutout-side protruding wall portion, is provided to be a part of the upper surface of the ring **122**. That is, the formation height of the upper surface of the wall portion **12c** coincides with the formation height of the upper surface of the ring **122**. Therefore, the groove **2n** as the gas escape flow path can be formed below the wall portion **12c** having a sufficient thickness.

[0239] Furthermore, the wall portion **12c**, which is the cutout-side protruding wall portion, faces the SiC wafer **24** in the semiconductor wafer mounting state and is disposed so as to fill a part of the notch **24n** without coming into contact with the SiC wafer **24**, whereby the amount of the source gas G**1** flowing through the groove **2n** via the notch **24n** can be limited not to exceed an allowable upper limit amount. This is because if the source gas G**1** exceeding the allowable upper limit amount flows into the groove **2n** via the notch **24n**, the source gas G**1** supplied to the front surface of the SiC wafer **24** may fall below an appropriate amount.

[0240] Therefore, the susceptor **21** of the second preferred embodiment can supply an appropriate amount of the source gas G**1** to the front surface of the SiC wafer **24** having the notch **24n**.

[0241] As described above, the susceptor **21**, which is the basic configuration of the second preferred embodiment, has effects similar to those of the susceptor **11**, which is the basic configuration of the first preferred embodiment, except that the mounting target is changed from the SiC wafer **14** to the SiC wafer **24**.

[0242] Also in the second preferred embodiment, modifications similar to the first and second modifications of the first preferred embodiment illustrated in FIGS. **6** to **13** can be configured.

[0243] That is, as a first modification, the front surface **13s** of the upper layer portion **13u** of the satellite disk **13** may be changed to a concave shape in which the center portion is recessed.

Furthermore, as a second modification, two auxiliary terrace portions may be provided between the terrace portion **12t** and the wall portion **12c** in plan view.

[0244] Note that, as illustrated in FIG. **17**, since the notch length L_n , which is the length of the notch **24n** in the radial direction, is shorter than the formation length of the orientation flat portion **14k**, it is assumed that the possibility that epi-crown occurs is relatively low.

[0245] In that case, it is conceivable that the epi-crown phenomenon can be suppressed only by not providing the wall portion **12c** and not forming the notch **24n** and the terrace portion **12t** in the vicinity thereof. When this possibility is satisfied, a configuration in which the formation of the wall portion **12c** is omitted is conceivable.

[0246] However, in a case where the wall portion **12c** is not formed in the ring **121**, when the SiC wafer **24** is mounted, it may be difficult to align the notch **24n** of the SiC wafer **24** with a terrace non-formation region where the terrace portion **12t** is not formed. Therefore, it is desirable to form the wall portion **12c** on the ring **121**. If the wall portion **12c** is not formed, the above-described difficulty of alignment can be reduced by ensuring a large terrace non-formation region.

[0247] In the susceptor **21**, which is the basic configuration of the second preferred embodiment, the terrace portion **12t** is supported from below on the front surface of the peripheral portion **13p** of the satellite disk **13**, but if there is no problem in strength, a plurality of partial terrace portions may be discretely provided instead of the terrace portion **12t** continuously formed as in the first preferred embodiment.

Third Preferred Embodiment

(Basic Configuration)

[0248] FIG. **18** is an explanatory view schematically illustrating a planar configuration of a susceptor **31**, which is a basic configuration of a third preferred embodiment of the present disclosure. FIG. **19** is an explanatory view schematically illustrating a configuration of an A3-A3 cross section of FIG. **18**, FIG. **20** is an explanatory view schematically illustrating a configuration of a B3-B3 cross section of FIG. **18**, and FIG. **21** is an explanatory view schematically illustrating a configuration of a C.sub.3-C.sub.3 cross section of FIG. **18**. In FIG. **18**, the planar structures of the ring **12** and the SiC wafer **34** as a mounting target, which are main components of the susceptor **31**, are schematically illustrated in an easily recognizable manner.

[0249] A mounting target of the susceptor **31**, which is the basic configuration of the wafer mounting apparatus according to the third preferred embodiment, is a SiC wafer **34**. In FIG. **18**, the SiC wafer **34** is formally illustrated in a circular shape in plan view, but the SiC wafer **34** illustrates the SiC wafer **14** or the SiC wafer **24**. That is, the mounting target of the susceptor **31** of the third preferred embodiment is the SiC wafer **14** having the orientation flat portion **14k** or the SiC wafer **24** having the notch **24n**.

[0250] Hereinafter, components similar to those of the susceptor **11** of the basic configuration of the first preferred embodiment illustrated in FIGS. **1** to **5** will be denoted by the same reference numerals, description thereof will be omitted as appropriate, and features of the susceptor **31** of the basic configuration of the third preferred embodiment will be mainly described.

[0251] Note that FIGS. **18** to **21** illustrate the ring **12** assuming a case where the SiC wafer **34** is the SiC wafer **14**. Therefore, when the SiC wafer **34** is the SiC wafer **24**, the ring **122** illustrated in FIGS. **14** to **16** is used instead of the ring **12**.

[0252] Hereinafter, for convenience of description, the susceptor **31** of the third preferred embodiment will be described assuming that the SiC wafer **34** is the SiC wafer **14**.

[0253] As illustrated in FIGS. **18** to **21**, the susceptor **31** includes, as main components, a combination of a satellite disk **13**, which is a disk having a circular shape in plan view and a protruding cross section, and a ring **12**.

[0254] The satellite disk **13**, which is a disk of the basic configuration, includes an upper layer portion **13u**, which is a protrusion, and includes a peripheral portion **13p** around the upper layer portion **13u**.

[0255] Three lift-up pins **25**, which are a plurality of lift-up members each having the same height, are mounted on the front surface **13s** of the upper layer portion **13u** of the satellite disk **13**. The three lift-up pins **25** hold the SiC wafer **34** by supporting the SiC wafer **34** from the back surface.

[0256] In the susceptor **31** of the third preferred embodiment, the front surface **13s** of the upper layer portion **13u** of the satellite disk **13** is a lift-up region.

[0257] Furthermore, an annular support member **17** is disposed below the outer peripheral portion of the back surface of the satellite disk **13** in plan view, and the satellite disk **13** is supported by the support member **17**.

[0258] As described above, the susceptor **31**, which is the basic configuration of the third preferred embodiment includes the ring **12**, the satellite disk **13**, the three lift-up pins **25**, and the support member **17** as main components.

[0259] In the susceptor **31** of the third preferred embodiment having such a configuration, the SiC wafer **34** can be held by mounting the SiC wafer **34** on the three lift-up pins **25** so that the SiC wafer **34** fits within the opening **O12** in plan view without being in contact with the ring **12**. This state is the semiconductor wafer mounting state related to the SiC wafer **34**.

[0260] When the SiC wafer **34** is mounted on the three lift-up pins **25**, a disk upper space **S13** is formed between the front surface **13s** of the upper layer portion **13u** of the satellite disk **13** and the back surface of the SiC wafer **34** except for a formation region of the three lift-up pins **25**.

[0261] Furthermore, a groove **2**, which is an annular groove having an annular shape along the outer periphery of the opening **O12** in plan view, is formed between the upper layer portion **13u** of the satellite disk **13** and the ring **12** on the back surface side of the SiC wafer **34**.

[0262] The annular groove **2** has one side surface as a side surface of the upper layer portion **13u** and the other side surface as the wall portion **12a**. Then, the groove **2** formed in the formation region of the terrace portion **12t** is a space above the terrace portion **12t** with the terrace portion **12t** as a bottom surface.

[0263] On the other hand, as illustrated in FIG. **19**, in the region where the terrace portion **12t** is not formed, the groove **2** immediately below the wall portion **12b** is a groove **2k**, and as illustrated in FIG. **21**, the groove **2** positioned immediately below both end peripheral regions of the wall portion **12b** is a groove **2m**.

[0264] The terrace portion **12t** of the ring **12** is provided so as not to have a contact relationship with the side surface of the upper layer portion **13u** of the satellite disk **13**. Therefore, an auxiliary groove **2X** communicating with the groove **2** in the depth direction can be provided between the front end of the terrace portion **12t** and the side surface of the upper layer portion **13u**.

[0265] When the susceptor **31** of the third preferred embodiment is applied to the semiconductor manufacturing apparatus **50** illustrated in FIG. **25**, the satellite disk **183** is replaced with the susceptor **31** including the ring **12**, the satellite disk **13**, the three lift-up pins **25**, and the support member **17**, and the SiC wafer **184** is replaced with the SiC wafer **34**.

[0266] The susceptor **31**, which is the basic configuration of the wafer mounting apparatus according to the third preferred embodiment of the present disclosure, achieves the semiconductor wafer mounting state related to the SiC wafer **34** by mounting the SiC wafer **34**, which is a semiconductor wafer, on the three lift-up pins **25**, which are a plurality of lift-up members. Therefore, the disk upper space **S13** can be provided between the back surface of the SiC wafer **34** and the front surface **13s** of the satellite disk **13**.

[0267] Furthermore, in the susceptor **31** of the third preferred embodiment, in the semiconductor wafer mounting state, the groove **2** which is an annular groove is formed between the upper layer portion **13u** of the satellite disk **13** and the ring **12** on the back surface side of the SiC wafer **34**.

[0268] As a result, the susceptor **31** according to the third preferred embodiment has an effect of being able to form an epitaxial layer having excellent in-plane uniformity on the front surface of the SiC wafer **34** in the semiconductor wafer mounting state, similarly to first and second preferred embodiments. Note that an epi thickness and a carrier concentration (impurity concentration) are

included as indices of in-plane uniformity.

[0269] As described above, the susceptor **31**, which is the basic configuration of the third preferred embodiment has effects similar to those of the susceptors **11** and **21**, which are the basic configurations of the first preferred embodiment and the second preferred embodiment, and further has the following unique effects.

[0270] The susceptor **31** of the third preferred embodiment has the front surface **13s** of the upper layer portion **13u** of the satellite disk **13** as a lift-up region, and can stably support the SiC wafer **34** by the three lift-up pins **25** mounted on the front surface **13s**.

[0271] Furthermore, in the susceptor **31** of the third preferred embodiment, the support member **17** that supports the satellite disk **13** from the back surface can suppress the phenomenon that the satellite disk **13** on which the SiC wafer **34** is mounted warps via the three lift-up pins **25**.

[0272] Also in the third preferred embodiment, modifications similar to the first and second modifications of the first preferred embodiment can be configured.

[0273] That is, as a first modification, the front surface **13s** of the upper layer portion **13u** of the satellite disk **13** may be changed to a concave shape in which the center portion is recessed.

However, in order to stably support the SiC wafer **34**, it is desirable that the region on which the three lift-up pins **25** are mounted is flat on the front surface **13s** of the upper layer portion **13u**.

[0274] Furthermore, as a second modification, two auxiliary terrace portions may be provided between the terrace portion **12t** and the wall portion **12b** (wall portion **12c**) in plan view.

Fourth Preferred Embodiment

(Basic Configuration)

[0275] FIG. **22** is an explanatory view schematically illustrating a planar configuration of a susceptor **41**, which is a basic configuration of the fourth preferred embodiment of the present disclosure. FIG. **23** is an explanatory view schematically illustrating a configuration of an A4-A4 cross section of FIG. **22**, and FIG. **24** is an explanatory view schematically illustrating a configuration of a B4-B4 cross section of FIG. **22**. Note that, in FIG. **22**, the planar structures of a ring **22** and a satellite disk **23**, which are main components of the susceptor **41**, are schematically illustrated in an easily recognizable manner.

[0276] A mounting target of the susceptor **41**, which is the basic configuration of the wafer mounting apparatus according to the fourth preferred embodiment, is the SiC wafer **14** having the orientation flat portion **14k**. That is, the susceptor **41** of the fourth preferred embodiment is a wafer mounting apparatus on which the SiC wafer **14** having the orientation flat portion **14k** is mounted.

[0277] Hereinafter, components similar to those of the susceptor **11** of the basic configuration of the first preferred embodiment illustrated in FIGS. **1** to **5** will be denoted by the same reference numerals, description thereof will be omitted as appropriate, and features of the susceptor **41** of the basic configuration of the fourth preferred embodiment will be mainly described.

[0278] As illustrated in FIGS. **22** to **24**, the susceptor **41** includes, as main components, a combination of the satellite disk **23**, which is a disk having a circular shape in plan view and a protruding cross section, and the ring **22**.

[0279] The ring **22** is provided with a wall portion **22a** and a wall portion **22b** serving as inner peripheral surfaces. That is, in the ring **22**, the wall portion **22b** is selectively provided as a part of the inner peripheral surface of the ring **22**, and the wall portion **22a** serving as a uniform inner peripheral surface is provided in a region where the wall portion **22b** is not provided.

[0280] The ring **22** has an opening O22 with the wall portion **22a** and the wall portion **22b** as outer peripheries. The opening O22 has a shape approximate to the wafer shape of the SiC wafer **14**, includes the SiC wafer **14** in plan view, and has a planar shape slightly larger than the SiC wafer **14**.

[0281] The wall portion **22b** of the ring **22** functions as a cutout-side protruding wall portion protruding toward a side of the opening O22 corresponding to the orientation flat portion **14k** which is a cutout portion of the SiC wafer **14**. As illustrated in FIG. **22**, the wall portion **22b** is

provided in parallel with the orientation flat portion **14k** in plan view, and a gap space **S22** in which a length in a lateral direction is a constant interval exists between the wall portion **22b** and the orientation flat portion **14k**.

[0282] The satellite disk **23**, which is a disk of the basic configuration, includes an upper layer portion **23u** which is a protrusion, includes an intermediate peripheral portion **23m** which is a first peripheral portion around the upper layer portion **23u**, and includes an outermost peripheral portion **23p** which is a second peripheral portion around the intermediate peripheral portion **23m**. In the satellite disk **23**, the formation height decreases in the order of a front surface **23s** of the upper layer portion **23u**, an intermediate peripheral portion front surface **23d** of the intermediate peripheral portion **23m**, and an outermost peripheral portion front surface **23e** of the outermost peripheral portion **23p**.

[0283] The ring **22** is mounted on the satellite disk **23** in such a manner that a bottom surface of the ring **22** is supported on the outermost peripheral portion front surface **23e** of the outermost peripheral portion **23p**, which is the second peripheral portion of the satellite disk **23**, along the entire circumference of the opening **O22**.

[0284] As illustrated in FIG. **23**, the wall portion **22b** serving as the cutout-side protruding wall portion does not have a contact relationship with the intermediate peripheral portion front surface **23d** of the intermediate peripheral portion **23m** of the satellite disk **23**.

[0285] In the susceptor **41** having the mechanism configuration of the fourth preferred embodiment, the front surface **23s** of the upper layer portion **23u** is a lift-up region for the three lift-up pins **25**.

[0286] The three lift-up pins **25** each having the same height are mounted on the front surface **23s** of the upper layer portion **23u** of the satellite disk **23**. The three lift-up pins **25** hold the SiC wafer **14** by supporting the SiC wafer **14** from the back surface.

[0287] Furthermore, an annular support member **27** is disposed below the outer peripheral portion of the back surface of the satellite disk **23** in plan view, and the satellite disk **23** is supported by the support member **27**.

[0288] As described above, the susceptor **41**, which is the basic configuration of the fourth preferred embodiment includes the ring **22**, the satellite disk **23**, the three lift-up pins **25**, and the support member **27** as main components.

[0289] In the susceptor **41** of the fourth preferred embodiment having such a configuration, the SiC wafer **14** can be held by mounting the SiC wafer **14** on the three lift-up pins **25** so that the SiC wafer **14** fits within the opening **O22** in plan view without being in contact with the ring **22**. This state is the semiconductor wafer mounting state related to the SiC wafer **14**.

[0290] During the semiconductor wafer mounting state related to the SiC wafer **14**, the wall portion **22b**, which is the cutout-side protruding wall portion, and the orientation flat portion **14k** are in a positional relationship of facing each other via the gap space **S22** at a constant interval.

[0291] When the SiC wafer **14** is mounted on the three lift-up pins **25**, a disk upper space **S23** is formed between the front surface **23s** of the upper layer portion **23u** of the satellite disk **23** and the back surface of the SiC wafer **14** except for a formation region of the three lift-up pins **25**.

[0292] Furthermore, a groove **2**, which is an annular groove having an annular shape along the outer periphery of the opening **O22** in plan view, is formed between the upper layer portion **23u** of the satellite disk **23** and the ring **12** on the back surface side of the SiC wafer **14**.

[0293] The annular groove **2** has one side surface as a side surface of the upper layer portion **23u** and the other side surface as a wall portion **12a**. As illustrated in FIG. **23**, the groove **2** immediately below the wall portion **12b** is a groove **2k**.

[0294] When the susceptor **41** of the fourth preferred embodiment is applied to the semiconductor manufacturing apparatus **50** illustrated in FIG. **25**, the satellite disk **183** is replaced with the susceptor **41** including the ring **22**, the satellite disk **23**, the three lift-up pins **25**, and the support member **27**, and the SiC wafer **184** is replaced with the SiC wafer **14**.

[0295] The susceptor **41**, which is the basic configuration of the wafer mounting apparatus according to the fourth preferred embodiment of the present disclosure, achieves the semiconductor wafer mounting state related to the SiC wafer **14** by mounting the SiC wafer **14** on the three lift-up pins **25**. Therefore, the disk upper space **S23** can be provided between the back surface of the SiC wafer **14** and the front surface **23s** of the satellite disk **23**.

[0296] Furthermore, in the susceptor **41** of the fourth preferred embodiment, the groove **2** which is an annular groove is formed between the upper layer portion **23u** of the satellite disk **23** and the ring **22** on the back surface side of the SiC wafer **14** during the semiconductor wafer mounting state.

[0297] As a result, the susceptor **41** according to the fourth preferred embodiment has an effect of being able to form an epitaxial layer having excellent in-plane uniformity on the front surface of the SiC wafer **14** in the semiconductor wafer mounting state, similarly to first to third preferred embodiments. Note that an epi thickness and a carrier concentration (impurity concentration) are included as indices of in-plane uniformity.

[0298] As described above, the susceptor **41**, which is the basic configuration of the fourth preferred embodiment, has effects similar to those of the susceptors **11**, **21**, and **31**, which are the basic configurations of the first to third preferred embodiments, and further has the following unique effects.

[0299] In the susceptor **41** of the fourth preferred embodiment, the ring **22** is mounted on the satellite disk **23** in such a manner that the bottom surface of the ring **22** is supported on the outermost peripheral portion front surface **23e** of the outermost peripheral portion **23p**, which is the second peripheral portion of the satellite disk **23**, along the entire circumference of the opening **O22**. Therefore, in the susceptor **41** of the fourth preferred embodiment, a combination structure of the satellite disk **23** and the ring **22** can be achieved without affecting the SiC wafer **14**.

[0300] Also in the fourth preferred embodiment, a modification similar to the first modification of the first preferred embodiment can be configured.

[0301] That is, as a first modification, the front surface **23s** of the upper layer portion **23u** of the satellite disk **23** may be changed to a concave shape in which the center portion is recessed. However, in order to stably support the SiC wafer **14**, it is desirable that the region on which the three lift-up pins **25** are mounted is flat on the front surface **23s** of the upper layer portion **23u**.

[0302] Further, the mounting target of the susceptor **41** of the fourth preferred embodiment is the SiC wafer **14**, but may be the SiC wafer **24** as in the susceptor **21** of the second preferred embodiment. In this case, in the ring **22**, a wall portion having a structure similar to the wall portion **12c** is provided instead of the wall portion **22b**.

<Application to Semiconductor Manufacturing Apparatus>

[0303] As described above, the semiconductor manufacturing apparatus **50** illustrated in FIG. **25** can be configured using the wafer mounting apparatus illustrated in the first to fourth preferred embodiments as a component.

[0304] When the susceptor **11** of the first preferred embodiment is applied to the semiconductor manufacturing apparatus **50**, the satellite disk **183** is replaced with the susceptor **11** including the ring **12**, the satellite disk **13**, and the three lift-up pins **15**, and the SiC wafer **184** is replaced with the SiC wafer **14**.

[0305] FIG. **26** is an explanatory view schematically illustrating a planar configuration of the semiconductor manufacturing apparatus **51**. The semiconductor manufacturing apparatus **51** is a novel semiconductor manufacturing apparatus in which the semiconductor manufacturing apparatus **50** illustrated in FIG. **25** is applied to the susceptor **11** of the first preferred embodiment.

[0306] As illustrated in the drawing, the planetary susceptor **182** accommodates eight susceptors **11** each having a ring **12**, a satellite disk **13**, and three lift-up pins **15** as main components. That is, in the upper layer portion of the planetary susceptor **182**, 8 susceptors **11** are arranged as 8 satellite disks **183** around the center part **187**.

[0307] Each of the eight susceptors **11** is mounted with the SiC wafer **14** and rotates along the disk rotation direction **R183**. In addition, the planetary susceptor **182** itself also rotates along the susceptor rotation direction **R182**.

[0308] Note that, when the susceptor **21** of the second preferred embodiment is applied to the semiconductor manufacturing apparatus **50**, the satellite disk **183** is replaced with the susceptor **21** including the ring **122**, the satellite disk **13**, and the three lift-up pins **15**, and the SiC wafer **184** is replaced with the SiC wafer **24**.

[0309] When the susceptor **31** of the third preferred embodiment is applied to the semiconductor manufacturing apparatus **50**, the satellite disk **183** is replaced with the susceptor **31** including the ring **12**, the satellite disk **13**, the three lift-up pins **25**, and the support member **17**, and the SiC wafer **184** is replaced with the SiC wafer **34**.

[0310] When the susceptor **41** of the fourth preferred embodiment is applied to the semiconductor manufacturing apparatus **50**, the satellite disk **183** is replaced with the susceptor **41** including the ring **22**, the satellite disk **23**, the three lift-up pins **25**, and the support member **27**, and the SiC wafer **184** is replaced with the SiC wafer **14**.

[0311] The semiconductor manufacturing apparatus **51** including any one of the wafer mounting apparatuses of the first to fourth preferred embodiments as a component is a novel semiconductor manufacturing apparatus. Note that both the SiC wafer **14** and the SiC wafer **24** are semiconductor wafers containing silicon carbide as a constituent material.

[0312] The semiconductor manufacturing apparatus **51** includes an induction heating coil **188** as a heating mechanism for heating a semiconductor wafer in the semiconductor wafer mounting state.

[0313] The novel semiconductor manufacturing apparatus **51** can form an epitaxial layer containing silicon carbide on the front surface of the SiC wafer **14** with good in-plane uniformity by supplying the source gas **G1** for epitaxial growth above the front surface of the SiC wafer **14** in a state where the SiC wafer **14** in the semiconductor wafer mounting state is heated by the induction heating coil **188** as a heating mechanism. Note that an epi thickness and a carrier concentration (impurity concentration) are included as indices of in-plane uniformity.

<Method of Manufacturing Semiconductor Device and Semiconductor Device Manufactured by Method of Manufacturing>

[0314] FIG. **27** is a flowchart illustrating a processing procedure of a method of manufacturing a semiconductor device using the novel semiconductor manufacturing apparatus **51**. Hereinafter, a method of manufacturing a semiconductor device using the semiconductor manufacturing apparatus **51** to which the susceptor **11** of the first preferred embodiment is applied will be described.

[0315] In step **ST1**, the SiC wafer **14** as a silicon carbide semiconductor substrate is mounted on the three lift-up pins **15** by the susceptor **11**.

[0316] That is, in step **ST1**, the SiC wafer **14** is mounted on the three lift-up pins **15** so that the SiC wafer **14** fits in the opening **O12** in plan view without being in contact with the ring **12**, thereby achieving the semiconductor wafer mounting state regarding the SiC wafer **14**.

[0317] In step **ST2**, the epitaxial growth process is performed on the front surface of the SiC wafer **14** in the semiconductor wafer mounting state. That is, the source gas **G1** for epitaxial growth is supplied to the front surface of the SiC wafer **14**, and the SiC wafer **14** is heated by the induction heating coil **188** as a heating mechanism.

[0318] As a result, an epitaxial layer is formed on the front surface of the SiC wafer **14** in the semiconductor wafer mounting state.

[0319] Thereafter, in step **ST3**, a semiconductor layer for element formation is formed in the epitaxial layer, and then in step **ST4**, a front surface structure for element formation is formed on the semiconductor layer for element formation.

[0320] As a result, a semiconductor element can be manufactured by the semiconductor layer and the front surface structure of the element formation.

[0321] FIG. 28 is a cross-sectional view illustrating a cross-sectional structure of a semiconductor device 100 as an example of a semiconductor device manufactured by the method of manufacturing the semiconductor device illustrated in FIG. 27.

[0322] As illustrated in the drawing, an N.sup.- epitaxial layer 62 is provided as an epitaxial layer on the front surface of the SiC substrate 61 corresponding to the SiC wafer 14. A P base layer 63 is selectively provided in an upper layer portion of the N.sup.- epitaxial layer 62, an N source layer 64 is selectively provided in a front surface of the P base layer 63, and a P.sup.+ contact layer 65 is provided between N source layers 64 and 64 in the front surface of the P base layer 63.

[0323] The gate electrode 67 is formed via a gate oxide film 66 between the pair of N source layers 64 and 64 formed in the different P base layers 63 and adjacent to each other. The gate electrode 67 is formed from above the N source layer 64 provided in the front surface of one P base layer 63 to above the one P base layer 63, above the N epitaxial layer 62, above the other P base layer 63, and above the N source layer 64 provided in the front surface of the other P base layer 63. An interlayer insulating film 68 is provided to cover the gate oxide film 66 and the gate electrode 67.

[0324] Such a combination of the gate oxide film 66, the gate electrode 67, and the interlayer insulating film 68 is formed at a plurality of locations. Therefore, the semiconductor device 100 includes a plurality of gate oxide films 66, a plurality of gate electrodes 67, and a plurality of interlayer insulating films 68. Then, a source electrode 69 is provided on the entire surface of the N.sup.- epitaxial layer 62 including the plurality of interlayer insulating films 68, and a drain electrode 70 is provided on a back surface of the SiC substrate 61.

[0325] Hereinafter, a process of manufacturing the semiconductor device 100 illustrated in FIG. 28 by the method of manufacturing the semiconductor device illustrated in FIG. 27 will be described.

[0326] The processing in step ST1 is processing of mounting the SiC wafer 14 to be the SiC substrate 61 finally on the susceptor 11.

[0327] The processing in step ST2 is processing of supplying the source gas G1 for epitaxial growth onto the front surface of the SiC wafer 14 mounted on the semiconductor manufacturing apparatus 51 and heating the SiC wafer 14 by the induction heating coil 188.

[0328] As a result, the epitaxial growth process is performed on the front surface of the SiC substrate 61, and the N.sup.- epitaxial layer 62 to be an epitaxial layer is formed on the front surface of the SiC substrate 61. Here, the concentration of the n-type impurity contained in the SiC substrate 61 and the N.sup.- epitaxial layer 62 is appropriately selected according to the withstand voltage of the semiconductor device to be manufactured.

[0329] Next, a semiconductor layer for element formation is formed in the N.sup.- epitaxial layer 62. The P base layer 63, the N source layer 64, and the P.sup.+ contact layer 65 can each be formed by implanting ions of p-type impurities and n-type impurities into the N.sup.- epitaxial layer 62 and diffusing the implanted ions in the N.sup.- epitaxial layer 62 by a heat treatment or the like.

[0330] The P base layer 63, the N source layer 64, and the P.sup.+ contact layer 65 serve as semiconductor layers for element formation. That is, the process of step ST3 is a process of forming the P base layer 63, the N source layer 64, and the P.sup.+ contact layer 65.

[0331] Next, the gate oxide film 66, the gate electrode 67, and the interlayer insulating film 68 are selectively formed on the front surface of the N.sup.- epitaxial layer 62 including the P base layer 63, the N source layer 64, and the P.sup.+ contact layer 65.

[0332] First, the SiC wafer 14 is heated in an atmosphere containing oxygen to form the gate oxide film 66 formed by SiO.sub.2 on the upper surface of the semiconductor layer for element formation. Then, polysilicon doped with n-type or p-type impurities is deposited on the gate oxide film 66 using a chemical vapor deposition (CVD) method or the like to form the gate electrode 67.

[0333] Thereafter, the interlayer insulating film 68 is formed on the gate electrode 67. A constituent material of the interlayer insulating film 68 is, for example, SiO.sub.2 or tetraethyl orthosilicate (TEOS). Through such steps, the plurality of gate oxide films 66, the plurality of gate electrodes 67, and the plurality of source electrodes 69 are selectively formed.

[0334] Then, a mask process is performed on the upper surface of the interlayer insulating film **68**, and a part of the interlayer insulating film **68** is etched through the opening of the resist to form a contact hole **59** penetrating the interlayer insulating film **68**.

[0335] The contact hole **59** is formed on the front surfaces of the N source layer **64** and the P.sup.+ contact layer **65**.

[0336] Next, a barrier metal (not illustrated) is formed in the contact hole **59** of the interlayer insulating film **68** and on the interlayer insulating film **68**. The barrier metal is formed by depositing Ti, TiN, or the like by a physical vapor deposition (PVD) method or a CVD method.

[0337] Next, the source electrode **69** is formed on the barrier metal. The source electrode **69** can be formed by, for example, depositing an aluminum silicon alloy (Al—Si-based alloy) on a barrier metal using a PVD method such as sputtering or vapor deposition.

[0338] Next, the drain electrode **70** is formed on the back surface of the SiC substrate **61**. The drain electrode **70** is formed by depositing an aluminum silicon alloy, titanium, or the like using a PVD method such as sputtering or vapor deposition, for example. In addition, the drain electrode **70** may be formed by laminating a plurality of metals such as an aluminum silicon alloy, titanium, nickel, or gold.

[0339] The front surface structure for element formation is the gate oxide film **66**, the gate electrode **67**, the interlayer insulating film **68**, the source electrode **69**, and the drain electrode **70**. That is, the processing of step ST4 is processing of forming the gate oxide film **66**, the gate electrode **67**, the interlayer insulating film **68**, the source electrode **69**, and the drain electrode **70**.

[0340] Note that, since a plurality of semiconductor devices **100**, which are silicon carbide semiconductor devices, is produced in a matrix on one SiC wafer **14**, one SiC wafer **14** is cut into individual semiconductor devices **100** by laser dicing or blade dicing to be formed into chips, and the semiconductor device **100** illustrated in FIG. **28** is completed.

[0341] As a result, it is possible to manufacture the semiconductor device **100** including a Metal-Oxide-Semiconductor Field Effect Transistor (NMOSFET) as a semiconductor element configured by a semiconductor layer for element formation and a front surface structure.

[0342] As described above, in the semiconductor device **100** manufactured by the method of manufacturing a semiconductor device using the semiconductor manufacturing apparatus **51** including the planetary susceptor **182**, the N.sup.— epitaxial layer **62** has high in-plane uniformity such as an epi thickness, and it is possible to suppress a structure having poor electrical characteristics in a step after the formation of the N.sup.— epitaxial layer **62**, so that productivity is improved.

[0343] In addition, since it is possible to suppress the occurrence of poor electrical characteristics when a semiconductor layer for element formation is formed by ion implantation or the like, it is possible to obtain the semiconductor device **100** having a highly reliable NMOSFET.

[0344] The method of manufacturing the semiconductor device illustrated in FIG. **27** includes at least the following steps (a) to (c). [0345] Step (a) is a step of mounting the SiC wafer **14** (SiC substrate **61**) on the three lift-up pins **15** so that the SiC wafer **14** fits in the opening O12 in plan view without being in contact with the ring **12**, and achieving the semiconductor wafer mounting state related to the SiC wafer **14**. [0346] Step (b) is a step of supplying the source gas G1 for epitaxial growth to the front surface of the SiC wafer **14** in the semiconductor wafer mounting state. [0347] Step (c) is a step of heating the SiC wafer **14** in the semiconductor wafer mounting state by the induction heating coil **188** as a heating mechanism.

[0348] Step (a) corresponds to step ST1 illustrated in FIG. **27**, and steps (b) and (c) correspond to step ST2 illustrated in FIG. **27**.

[0349] Therefore, after the execution of the above-described steps (a) to (c), the N.sup.— epitaxial layer **62** illustrated in FIG. **28** is formed on the front surface of the SiC substrate **61** corresponding to the semiconductor wafer.

[0350] In the method of manufacturing a semiconductor device using the novel semiconductor

manufacturing apparatus **51**, by executing steps (a) to (c), the N.sup.- epitaxial layer **62** to be an epitaxial layer containing silicon carbide can be formed with good in-plane uniformity on the front surface of the SiC wafer **14** in the semiconductor wafer mounting state. Note that an epi thickness and a carrier concentration (impurity concentration) are included as indices of in-plane uniformity. [0351] That is, the in-plane uniformity of the N.sup.- epitaxial layer **62** formed on the SiC wafer **14** in the semiconductor wafer mounting state can be enhanced by the semiconductor device manufacturing method using the novel semiconductor manufacturing apparatus **51**.

[0352] In the semiconductor device **100** that is a semiconductor device manufactured by the method of manufacturing a semiconductor device using the novel semiconductor manufacturing apparatus **51**, since the semiconductor layer for element formation is formed in the N.sup.- epitaxial layer **62** layer having high in-plane uniformity, it is possible to improve the reliability of the NMOSFET that is a semiconductor element including the semiconductor layer for element formation.

[0353] Note that, in the present disclosure, each preferred embodiment can be freely combined, and each preferred embodiment can be appropriately modified or omitted within the scope of the disclosure.

[0354] Hereinafter, various aspects of the present disclosure will be collectively described as appendices.

Appendix 1

[0355] A wafer mounting apparatus that mounts a semiconductor wafer having a cutout portion, the wafer mounting apparatus including: [0356] a disk having a protruding upper layer portion; [0357] a ring disposed on the disk and having an opening in a center; [0358] a plurality of lift-up members that is arranged on a lift-up region provided on the disk or the ring and that supports the semiconductor wafer from a back surface, in which [0359] the ring has a cutout-side protruding wall portion protruding toward the opening, and [0360] in a semiconductor wafer mounting state in which the semiconductor wafer is mounted on the plurality of lift-up members so that the semiconductor wafer fits in the opening in plan view without being in contact with the ring, [0361] the cutout-side protruding wall portion faces the cutout portion of the semiconductor wafer, a disk upper space is formed between a front surface of the upper layer portion of the disk and a back surface of the semiconductor wafer except for a formation region of the plurality of lift-up members, an annular groove is formed along an outer periphery of the opening in plan view between the upper layer portion of the disk and the ring on a side of the back surface of the semiconductor wafer, and a side surface of the upper layer portion of the disk is a side surface of the annular groove.

Appendix 2

[0362] The wafer mounting apparatus according to appendix 1, in which [0363] the disk has a peripheral portion around the upper layer portion, a front surface of the peripheral portion has a formation height lower than a front surface of the upper layer portion, and a step exists from the front surface of the peripheral portion to the front surface of the upper layer portion, [0364] the ring has a terrace portion protruding toward a side of the opening, and the terrace portion does not overlap the cutout-side protruding wall portion in plan view, [0365] the ring is mounted on the disk in such a manner that the terrace portion is supported on a front surface of the peripheral portion of the disk, and [0366] a thickness of the terrace portion in a height direction is thinner than the step on the disk.

Appendix 3

[0367] The wafer mounting apparatus according to appendix 2, in which [0368] the plurality of lift-up members is mounted on a front surface of the terrace portion of the ring, and [0369] the lift-up region includes a front surface region of the terrace portion.

Appendix 4

[0370] The wafer mounting apparatus according to appendix 2 or 3, in which [0371] the terrace

portion does not have a contact relationship with a side surface of the upper layer portion.

Appendix 5

[0372] The wafer mounting apparatus according to any one of appendices 2 to 4, in which [0373] the opening is shaped to include the semiconductor wafer and has a larger formation area than the semiconductor wafer, and [0374] the cutout-side protruding wall portion has no contact relationship with a front surface of the peripheral portion of the disk.

Appendix 6

[0375] The wafer mounting apparatus according to appendix 5, in which [0376] a formation height of a lower surface of the cutout-side protruding wall portion is set to be equal to or higher than a formation height of upper surfaces of the plurality of lift-up members.

Appendix 7

[0377] The wafer mounting apparatus according to appendix 5 or 6, in which [0378] an upper surface of the cutout-side protruding wall portion is a part of an upper surface of the ring.

Appendix 8

[0379] The wafer mounting apparatus according to any one of appendices 1 to 7, in which [0380] the upper layer portion of the disk has a recessed shape in which a center portion is recessed.

Appendix 9

[0381] The wafer mounting apparatus according to appendix 3, in which [0382] the ring further includes an auxiliary terrace portion protruding toward the side of the opening in addition to the terrace portion, and the auxiliary terrace portion is provided between the terrace portion and the cutout-side protruding wall portion in plan view, and has a thickness in a height direction smaller than that the thickness of the terrace portion, and [0383] the ring is mounted on the disk in a mode in which the terrace portion and the auxiliary terrace portion are supported on a front surface of the peripheral portion of the disk.

Appendix 10

[0384] The wafer mounting apparatus according to appendix 2, in which [0385] the plurality of lift-up members is mounted on the upper layer portion of the disk, and [0386] the lift-up region includes a front surface region of the upper layer portion.

Appendix 11

[0387] The wafer mounting apparatus according to appendix 10, further including: [0388] a support member that supports the disk from a back surface.

Appendix 12

[0389] The wafer mounting apparatus according to appendix 1, in which [0390] the disk includes a first peripheral portion around the upper layer portion and a second peripheral portion around the first peripheral portion, and a formation height decreases in order of a front surface of the upper layer portion, a front surface of the first peripheral portion, and a front surface of the second peripheral portion, and [0391] the ring is mounted on the disk in such a manner that a bottom surface of the ring is supported on the front surface of the second peripheral portion of the disk along an entire circumference of the opening.

Appendix 13

[0392] The wafer mounting apparatus according to any one of appendices 1 to 12, in which [0393] the cutout portion includes an orientation flat portion, and [0394] the cutout-side protruding wall portion is disposed to face the orientation flat portion via a gap at a constant interval in the semiconductor wafer mounting state.

Appendix 14

[0395] The wafer mounting apparatus according to any one of appendices 1 to 12, in which [0396] the cutout portion includes a notch, and [0397] the cutout-side protruding wall portion is disposed so as to fill a part of the notch without coming into contact with the semiconductor wafer in the semiconductor wafer mounting state.

Appendix 15

[0398] A semiconductor manufacturing apparatus including the wafer mounting apparatus according to any one of appendices 1 to 14, in which [0399] the semiconductor wafer contains silicon carbide as a constituent material, and [0400] the wafer mounting apparatus further includes: [0401] a heating mechanism that heats the semiconductor wafer in the semiconductor wafer mounting state.

Appendix 16

[0402] A method of manufacturing a semiconductor device using the wafer mounting apparatus according to any one of appendices 1 to 14, in which [0403] the semiconductor wafer contains silicon carbide as a constituent material, [0404] the wafer mounting apparatus further includes a heating mechanism that heats the semiconductor wafer, [0405] the method includes the steps of: [0406] (a) achieving the semiconductor wafer mounting state by mounting the semiconductor wafer on the plurality of lift-up members so that the semiconductor wafer fits in the opening in plan view without being in contact with the ring; [0407] (b) supplying a source gas for epitaxial growth to a front surface of the semiconductor wafer in the semiconductor wafer mounting state; and [0408] (c) heating the semiconductor wafer in the semiconductor wafer mounting state by the heating mechanism, and [0409] after the steps (a) to (c) are performed, an epitaxial layer is formed on a front surface of the semiconductor wafer in the semiconductor wafer mounting state.

Appendix 17

[0410] A semiconductor device including: [0411] the epitaxial layer formed by the method of manufacturing a semiconductor device according to appendix 16; and [0412] a semiconductor layer for element formation formed in the epitaxial layer.

[0413] While the disclosure has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous modifications and variations can be devised.

Claims

1. A wafer mounting apparatus that mounts a semiconductor wafer having a cutout portion, the wafer mounting apparatus comprising: a disk having a protruding upper layer portion; a ring disposed on the disk and having an opening in a center; a plurality of lift-up members that is arranged on a lift-up region provided on the disk or the ring and that supports the semiconductor wafer from a back surface, wherein the ring has a cutout-side protruding wall portion protruding toward the opening, and in a semiconductor wafer mounting state in which the semiconductor wafer is mounted on the plurality of lift-up members so that the semiconductor wafer fits in the opening in plan view without being in contact with the ring, the cutout-side protruding wall portion faces the cutout portion of the semiconductor wafer, a disk upper space is formed between a front surface of the upper layer portion of the disk and a back surface of the semiconductor wafer except for a formation region of the plurality of lift-up members, an annular groove having an annular shape is formed along an outer periphery of the opening in plan view between the upper layer portion of the disk and the ring on a side of the back surface of the semiconductor wafer, and a side surface of the upper layer portion of the disk is a side surface of the annular groove.
2. The wafer mounting apparatus according to claim 1, wherein the disk has a peripheral portion around the upper layer portion, a front surface of the peripheral portion has a formation height lower than a front surface of the upper layer portion, and a step exists from the front surface of the peripheral portion to the front surface of the upper layer portion, the ring has a terrace portion protruding toward a side of the opening, and the terrace portion does not overlap the cutout-side protruding wall portion in plan view, the ring is mounted on the disk in such a manner that the terrace portion is supported on a front surface of the peripheral portion of the disk, and a thickness of the terrace portion in a height direction is thinner than the step on the disk.
3. The wafer mounting apparatus according to claim 2, wherein the plurality of lift-up members is

mounted on a front surface of the terrace portion of the ring, and the lift-up region includes a front surface region of the terrace portion.

4. The wafer mounting apparatus according to claim 2, wherein the terrace portion does not have a contact relationship with a side surface of the upper layer portion.

5. The wafer mounting apparatus according to claim 2, wherein the opening is shaped to include the semiconductor wafer and has a larger formation area than the semiconductor wafer, and the cutout-side protruding wall portion has no contact relationship with a front surface of the peripheral portion of the disk.

6. The wafer mounting apparatus according to claim 5, wherein a formation height of a lower surface of the cutout-side protruding wall portion is set to be equal to or higher than a formation height of upper surfaces of the plurality of lift-up members.

7. The wafer mounting apparatus according to claim 5, wherein an upper surface of the cutout-side protruding wall portion is a part of an upper surface of the ring.

8. The wafer mounting apparatus according to claim 1, wherein the upper layer portion of the disk has a recessed shape in which a center portion is recessed.

9. The wafer mounting apparatus according to claim 3, wherein the ring further includes an auxiliary terrace portion protruding toward the side of the opening in addition to the terrace portion, and the auxiliary terrace portion is provided between the terrace portion and the cutout-side protruding wall portion in plan view, and has a thickness in a height direction smaller than that the thickness of the terrace portion, and the ring is mounted on the disk in a mode in which the terrace portion and the auxiliary terrace portion are supported on a front surface of the peripheral portion of the disk.

10. The wafer mounting apparatus according to claim 2, wherein the plurality of lift-up members is mounted on the upper layer portion of the disk, and the lift-up region includes a front surface region of the upper layer portion.

11. The wafer mounting apparatus according to claim 10, further comprising: a support member that supports the disk from a back surface.

12. The wafer mounting apparatus according to claim 1, wherein the disk includes a first peripheral portion around the upper layer portion and a second peripheral portion around the first peripheral portion, and a formation height decreases in order of a front surface of the upper layer portion, a front surface of the first peripheral portion, and a front surface of the second peripheral portion, and the ring is mounted on the disk in such a manner that a bottom surface of the ring is supported on the front surface of the second peripheral portion of the disk along an entire circumference of the opening.

13. The wafer mounting apparatus according to claim 1, wherein the cutout portion includes an orientation flat portion, and the cutout-side protruding wall portion is disposed to face the orientation flat portion via a gap at a constant interval in the semiconductor wafer mounting state.

14. The wafer mounting apparatus according to claim 1, wherein the cutout portion includes a notch, and the cutout-side protruding wall portion is disposed so as to fill a part of the notch without coming into contact with the semiconductor wafer in the semiconductor wafer mounting state.

15. A semiconductor manufacturing apparatus comprising the wafer mounting apparatus according to claim 1, wherein the semiconductor wafer contains silicon carbide as a constituent material, and the wafer mounting apparatus further includes a heating mechanism that heats the semiconductor wafer in the semiconductor wafer mounting state.

16. A method of manufacturing a semiconductor device using the wafer mounting apparatus according to claim 1, wherein the semiconductor wafer contains silicon carbide as a constituent material, the wafer mounting apparatus further includes a heating mechanism that heats the semiconductor wafer, the method comprises the steps of: (a) achieving the semiconductor wafer mounting state by mounting the semiconductor wafer on the plurality of lift-up members so that the

semiconductor wafer fits in the opening in plan view without being in contact with the ring; (b) supplying a source gas for epitaxial growth to a front surface of the semiconductor wafer in the semiconductor wafer mounting state; and (c) heating the semiconductor wafer in the semiconductor wafer mounting state by the heating mechanism, and after the steps (a) to (c) are performed, an epitaxial layer is formed on a front surface of the semiconductor wafer in the semiconductor wafer mounting state.

17. A semiconductor device comprising: the epitaxial layer formed by the method of manufacturing a semiconductor device according to claim **16**; and a semiconductor layer for element formation formed in the epitaxial layer.
